EdQuantum

Portable Quantum Experiments

Lab Book

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LAB BOOK

First Edition
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Laser Safety

About This Laser

The laser used in these labs is a red diode laser operating at approximately 650 nm. It is a Class II device under the ANSI Z136.1 standard and the IEC 60825-1 classification. Class II lasers emit visible light at 1 mW or less. The human blink reflex - approximately 0.25 seconds - is considered sufficient to prevent eye injury from momentary, accidental exposure. However, deliberate staring into the beam or its specular reflections is still a hazard. Treat this laser with the same respect you would give any laboratory instrument that can cause irreversible injury.

Required Eyewear

Class II laser safety eyewear rated for 650 nm wavelength is required during all lab sessions when the laser is energized. Instructors are responsible for sourcing eyewear with an appropriate Optical Density (OD) for the laser power in use. At 650 nm and Class II power levels, eyewear rated OD 1 or greater is generally sufficient - confirm with your institution's laser safety officer or eyewear vendor for your specific device. Eyewear must be worn by all persons within the Nominal Hazard Zone (NHZ), not just the student operating the laser.

Classroom Protocols - Based on ANSI Z136.1 Educational Guidelines

1. Designated Laser Operator (DLO)

Each lab session must have one Designated Laser Operator. The DLO is responsible for: turning the laser on and off, confirming that all persons in the NHZ are wearing appropriate eyewear before energizing, and calling a laser stop if any unsafe condition arises. In a two-student kit, the DLO role rotates each lab - see the schedule in each Instructor Page. For solo students, the instructor holds the DLO role permanently.

2. Nominal Hazard Zone (NHZ)

The NHZ is the region within which direct, reflected, or scattered laser radiation exceeds the Maximum Permissible Exposure (MPE). For this Class II 650 nm setup on a bench, treat the entire optical breadboard and a 30 cm margin around it as the NHZ. No person's eyes should enter this zone without eyewear during laser operation. Post a visible boundary (tape on the bench surface is sufficient) if multiple groups are working in close proximity.

3. Beam Stop Procedure

Before adding, removing, or adjusting any optical component - polarizer, wave plate, beamsplitter, or mount - the beam must be blocked. Do not rely on switching the laser off alone; a beam block (an opaque card placed in the beam path) provides a visible, physical confirmation that the beam is stopped. The sequence is: (1) place beam block, (2) make the change, (3) confirm eyewear, (4) remove beam block, (5) observe. This sequence must be followed every time, without exception.

4. Posted Warning

A laser-in-use warning notice must be posted on the laboratory door whenever the laser is energized. The notice should include the laser class (Class II), wavelength (650 nm), and the name of the instructor or DLO responsible for the session. Remove the notice when the laser is secured at the end of the session.

5. No Specular Surfaces Rule

Jewelry, watches, rings, and any reflective tools must be removed from hands and wrists before working near the beam path. Unexpected specular reflections from metal surfaces can redirect the beam unpredictably. Instruct students to check their hands before beginning assembly each session.

6. First-Session Safety Briefing

Before Lab 1, the instructor must deliver a verbal safety briefing covering all five points above. Students should demonstrate understanding by correctly identifying: (a) who the DLO is for the first session, (b) where the NHZ boundary is on their bench, and (c) the beam stop procedure before adjusting any component. This briefing does not need to be repeated in full for subsequent labs, but the DLO assignment must be confirmed at the start of every session.

7. Incident Protocol

If a student receives any direct beam exposure to the eye - even briefly - the laser must be secured immediately and the student must be evaluated by a medical professional. Do not wait for symptoms; laser eye injuries are not always immediately apparent. Report the incident to your institution's environmental health and safety office according to your local protocols.

How to Use the Visual Brightness Scale

In these labs you will observe laser light at two output ports of a polarizing beamsplitter. Instead of a light meter, you will use your eyes and a simple five-level scale to describe what you see. Every observation table in this book uses the same scale.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

What each level means

Level	What you see on the white screen
NONE	No visible spot at all. The screen looks the same as with the laser off.
LITTLE	A faint glow is visible. You have to look carefully to see it.
HALF	A clearly visible spot, noticeably dimmer than the full beam - roughly halfway between nothing and full brightness.
MOST	A bright spot. Close to full brightness but slightly reduced - clearly not all the light.
ALL	The full beam. This is your reference brightness, set at the start of each lab with only the laser and linear polarizer in the beam path.

Setting your 'ALL' reference - do this at the start of every lab

Before placing any wave plates or beamsplitters, set up only the laser and the linear polarizer. Rotate the polarizer to maximum brightness. That brightness is your ALL for this lab session. Every subsequent observation is relative to this moment. Why? Because room lighting, laser batteries, and your eyes all vary. Anchoring to a physical reference in the room makes your observations consistent and honest.

Tips for good observations

- Hold the white observation screen flat and perpendicular to the beam.
- Give your eyes a moment to adjust when moving between bright and dim setups.

- When judging HALF: cover one port, note the brightness, then cover the other. If they look similar, call it HALF / HALF.
- If two students disagree on a level, record both and discuss - that disagreement is useful data.

Optical Board Coordinate System

The optical board uses a simple letter-number coordinate system. Columns are labeled A through W, left to right. Rows are labeled 1 through 11, bottom to top. The origin is at the bottom-left corner of the board.

Every component position in this book is written as Column [letter], Row [number] - for example, Column B, Row 6. Row 6 is the standard beam path row used in all eight labs. All components travel along this row so the laser beam stays aligned across the full length of the board.

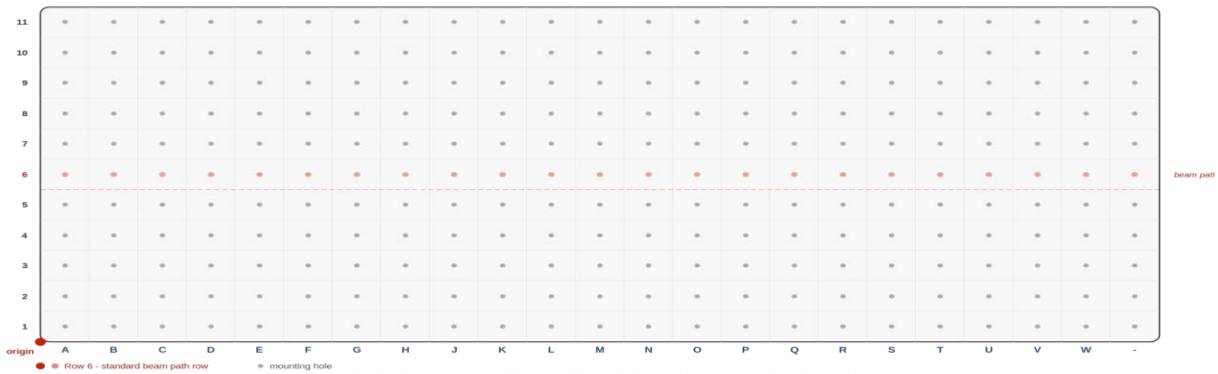

Fig C.1 - Optical board coordinate system. Columns A-W left to right. Rows 1-11 bottom to top. Origin at bottom-left corner.

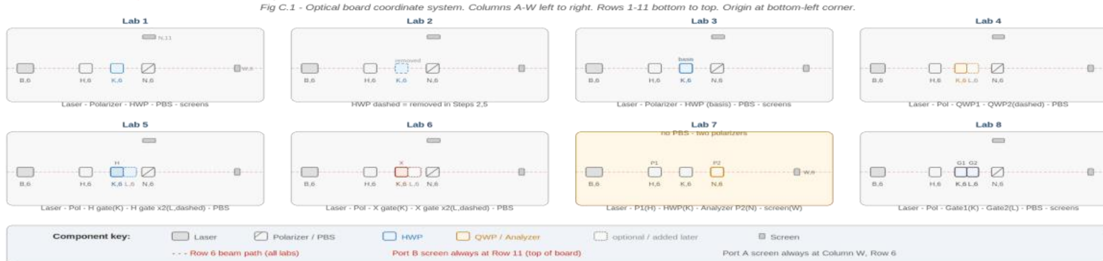

Fig C.2 - Component layout for all 8 labs. All components travel along Row 6 (beam path row). Port B screen at Column N, Row 11. Port A screen at Column W, Row 6.

How to read a coordinate

Column B, Row 6 means: count two columns from the left edge (Column B), then find the row that is six holes up from the bottom edge (Row 6). That hole is where the component post is seated.

Port B screens always sit at Column N, Row 11 - directly above the PBS at the top of the board. Port A screens always sit at Column W, Row 6 - at the right edge of the beam path.

LAB	Introduction to Polarization & Qubit Encoding

1

Goal

- Demonstrate how a half-wave plate controls the polarization state of laser light to encode binary qubit states $|0\rangle$ and $|1\rangle$ by routing light to Port A (horizontal) or Port B (vertical) of a polarizing beamsplitter.
- Observe the equal superposition state ($|0\rangle + |1\rangle$) by rotating the half-wave plate to 22.5° and confirming approximately equal brightness at both PBS output ports.
- Apply qubit encoding in a practical communication exercise: transmit and decode five binary signals using polarization states, then check for errors.

Theory

In this lab we introduce the relationship between photon polarization and qubit encoding. A qubit is a unit of information that can be in the '$|0\rangle$' state, the '$|1\rangle$' state, or a mixture of both at once. Picture a qubit like a coin: heads, tails, or spinning in the air - both and neither until you look. Horizontal polarization is $|0\rangle$; vertical polarization is $|1\rangle$.

For light to become polarized, it must pass through a linear polarizer - a device that transmits waves vibrating in one direction and blocks the rest (think of a picket fence that only lets through waves aligned with the slats). A half-wave plate (HWP) then rotates that polarization without absorbing the light, switching it between states. A polarizing beamsplitter (PBS) reads the result: horizontal light goes to Port A ($|0\rangle$), vertical light reflects to Port B ($|1\rangle$).

When the HWP sits at 22.5°, the polarization becomes a superposition - neither purely horizontal nor vertical. The PBS cannot choose a side; it splits the beam roughly equally. That equal split is the optical signature of superposition.

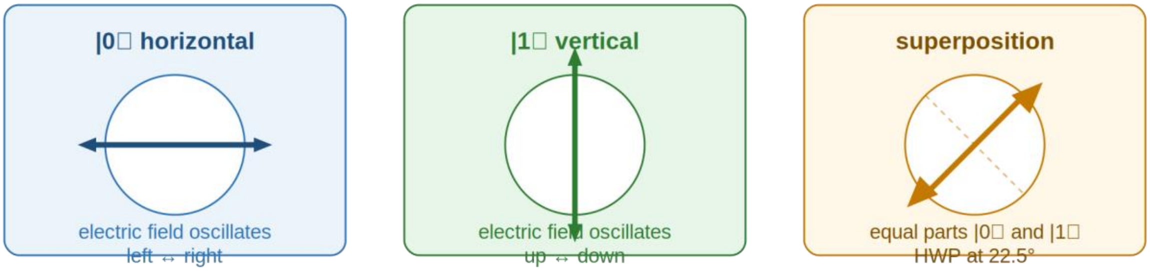

Fig 1.2 — Polarization states as qubit encodings. Each circle is a cross-section of the beam. The arrow shows the electric field direction.

Equipment

- Laser (650 nm red diode, on optical mount)
- Linear polarizer (always in beam path)
- Half-wave plate (HWP) - zero-order, 650 nm
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
Half-wave plate (HWP)	Column K, Row 6
Polarizing beamsplitter (PBS)	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

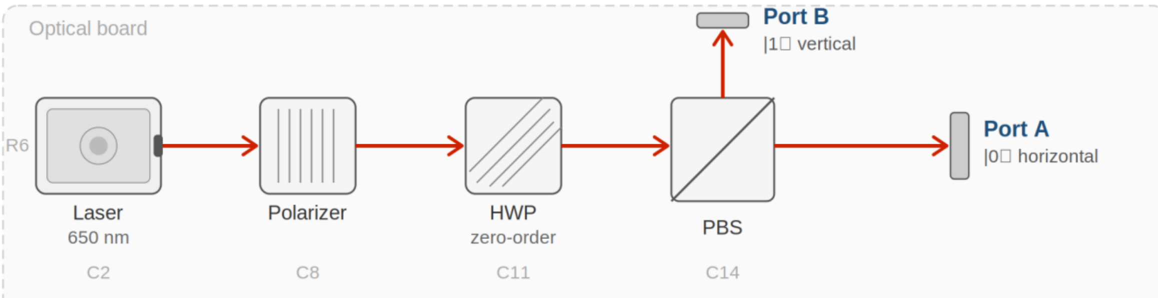

Fig 1.1 — Lab 1 beam path (top-down view). Red line = laser beam. Screen shown in gray.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Place the laser at Column B, Row 6 and the linear polarizer at Column H, Row 6. Rotate the polarizer until the beam on the white screen is as bright as it will go. This is your ALL for today.

Brightness through polarizer only (should be ALL): ____

Step 2 - Add the HWP and PBS

Place the HWP at Column K, Row 6. Place the PBS at Column N, Row 6. Position screens at Port A (Column W, Row 6) and Port B (Column N, Row 11). Rotate the HWP until Port A is as bright as possible and Port B is as dark as possible.

Port A brightness: ____

Port B brightness: ____

Step 3 - Flip to $|1\rangle$

Rotate the HWP exactly 45° from its current position.

Port A brightness: ____

Port B brightness:

____

____

LQ1) Does Port B now show ALL or close to it? Where did the light go, and what did the HWP do to cause this?

Answer:

____

____

____

Step 4 - Superposition at 22.5°

Rotate the HWP to 22.5° - halfway between the $|0\rangle$ and $|1\rangle$ settings.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Visual Observation (None / Little / Half / Most / All)
Port A ($|0\rangle$)	
Port B ($|1\rangle$)	

LQ2) Are both ports at roughly the same brightness? What does that tell you about the polarization state right now?

I expect both ports to show ___ because ___.

Answer:

____

____

____

Step 5 - Binary message exercise

Send a five-signal message using polarization. One person is the Encoder (sets the HWP angle); the other is the Decoder (watches ports). If working alone, set the angle, look away, then observe.

Encoder key: To send '0': HWP to 0° (light at Port A). To send '1': HWP to 45° (light at Port B).

Decoder key: Light at Port A = '0'. Light at Port B = '1'.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Signal #	Bit to Send	HWP Angle	Port Lit (A or B)	Decoder Reads	Match? (✓/✗)
1					
2					
3					
4					
5					

LQ3) Did any signals mismatch? If so, what do you think caused the error?

Answer:

____

____

____

LQ4) What would happen to the message if the HWP were set to 22.5° for every signal?

Think about what the PBS sees in superposition. I expect ___ because ___.

Answer:

____

____

____

Turn off the laser and take down the apparatus.

Post-Lab Questions

1. How does the half-wave plate change the direction of polarized light? How is it different from what the linear polarizer does?

____

____

____

____

2. Is the PBS creating the 1s and 0s, or sorting them based on what the HWP already did? Explain your reasoning.

____

____

____

____

3. A red laser was used. If you switched to a green laser, would the $|0\rangle$ and $|1\rangle$ HWP angles change? Why or why not?

____

____

____

____

4. A single photon is prepared in $|0\rangle$ and sent through the polarizer and HWP. How would its story differ from the laser beam you observed?

   Hint: think about what 'brightness at Port A' means for one photon versus many.

____

____

____

____

5. We used a laser with enormous numbers of photons. What part of this lab still behaves like single-qubit physics, and what part is only a classical average?

INSTRUCTOR PAGE - LAB 1: INTRODUCTION TO POLARIZATION & QUBIT ENCODING

NOT FOR STUDENT DISTRIBUTION

☐ **Designated Laser Operator (DLO) - Lab 1**

Lab 1 DLO: assign to Student A (or the more confident student in the pair). The DLO confirms eyewear before energizing, calls beam stops before any component is moved, and turns off the laser at the end. Rotate the DLO role to Student B for Lab 2.

Equipment Notes - Lab 1

Equipment note: This lab uses the 650 nm red diode laser and one zero-order HWP rated for 650 nm. Zero-order wave plates have significantly tighter angular tolerance than the multi-order plates listed in older BOMs. A tilt of even 5–10° from normal incidence can introduce measurable retardation error. Ensure the HWP is seated flat in the RSP05 rotation mount and the mount post is vertical. The PBS cube (Lambda PB-10B-650) is optimized for 650 nm - correct wavelength matching is important for the high extinction ratio students observe when encoding $|0\rangle$ and $|1\rangle$.

Equipment note: Zero-order vs. multi-order wave plates: A multi-order plate achieves the desired retardation by accumulating many full wavelength cycles plus the fractional retardation needed. This makes it sensitive to wavelength and temperature. A zero-order plate is designed so the net retardation is exactly the desired fraction (half or quarter wavelength) with no full-cycle accumulation. This gives better wavelength selectivity, lower sensitivity to temperature variation, and tighter angular tolerance. For classroom use at a fixed wavelength (650 nm), zero-order is the correct choice.

LQ Answer Key - Lab 1

LQ1 - Quick Reference: Yes - Port B shows ALL or MOST. The HWP rotated the polarization 90°, sending all light to the vertical output of the PBS.

Extended explanation: The HWP obeys the double-angle rule: a 45° plate rotation produces a 90° polarization rotation. The PBS transmits horizontal (Port A) and reflects vertical (Port B). Before the rotation, all light was horizontal → Port A. After the 45° HWP rotation, all light is vertical → Port B. No energy is absorbed; it is entirely redirected. Students sometimes expect the light to split between ports - this is a good moment to reinforce that a deterministic (non-superposition) input produces a deterministic output.

LQ2 - Quick Reference: Yes - both ports show approximately HALF. The HWP at 22.5° produces a 45° polarization rotation, creating equal H and V components.

Extended explanation: The double-angle rule gives 22.5° × 2 = 45° polarization rotation. At 45° from horizontal, the light has equal projections onto the PBS H and V axes: $\cos^2(45°) = \sin^2(45°) = 0.5$.

With many photons (a laser) this shows as equal brightness. For a single photon, the state $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 = |\beta|^2 = 0.5$ means a 50% probability of detection at each port - but only one photon arrives, so only one port fires. The brightness students see is the statistical average of many such single-photon events.

LQ3 - Quick Reference: Possible mismatches. Likely cause: the HWP was not fully rotated to 0° or 45°, or the PBS was slightly misaligned.

Extended explanation: In a real quantum channel with single photons and basis mismatch, errors are fundamental and unavoidable. Here, with a laser, all errors are classical (mechanical, alignment). This distinction is worth making explicitly: in this lab, errors are bugs; in a real QKD system, some errors are features - they signal eavesdropping.

LQ4 - Quick Reference: The decoder would record approximately half 0s and half 1s randomly - the message would be unreadable.

Extended explanation: At 22.5° the light is in superposition. The PBS randomly routes it to Port A or Port B on any given observation. With many photons the split is statistically even, but no individual signal carries determinate information. This is an important conceptual bridge: superposition is not useful for encoding classical bits - it is useful precisely because it is indeterminate, which is what makes QKD work.

Post-Lab Answer Key - Lab 1

Q1 - Quick Reference: The HWP rotates the polarization direction by twice its rotation angle, without absorbing light. The polarizer selects a single polarization direction by blocking all others.

Extended explanation: The polarizer is a dissipative element: photons with the wrong polarization are absorbed or deflected out of the beam. The HWP is unitary: it acts on all photons equally, simply changing the direction of oscillation. This is the classical analogue of the distinction between a projective measurement (polarizer) and a unitary gate (HWP). Instructors comfortable with linear algebra can note that the polarizer projects onto a subspace while the HWP applies a rotation matrix.

Q2 - Quick Reference: The PBS is sorting - the HWP created the state that the PBS then separates.

Extended explanation: The PBS has no 'knowledge' of what constitutes $|0\rangle$ or $|1\rangle$ - it simply separates horizontal from vertical. The HWP is the encoder. This is analogous to quantum computing: the gate (HWP) prepares the state; the measurement (PBS + observation) reads it. The PBS cannot change what state the qubit is in; it can only report what it finds.

Q3 - Quick Reference: No - the laser wavelength sets the HWP's optical behavior, but the |0⟩ and |1⟩ angles are defined by the geometry (0° = horizontal, 45° = vertical inversion), not the wavelength. The angles would stay the same; the plate would need to be designed for the new wavelength to maintain correct retardation.

Extended explanation: This is a subtle but important distinction. The angles 0° and 45° on the HWP are geometric definitions of our qubit basis - they are not wavelength-dependent. However, the retardation of the wave plate (whether it actually provides a half-wave shift) is wavelength-dependent. A plate optimized for 650 nm used with a 532 nm green laser would provide incorrect retardation, producing impure polarization states and degraded contrast at the PBS.

Q4 (single photon) - Quick Reference: With one photon, there is no brightness to measure. At Port A, either the photon arrives or it does not - a single binary event. The laser brightness is the average of many such events.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The key addition here: the HWP at 22.5° produces the state $(1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$. The Born rule gives $P(|0\rangle) = |1/\sqrt{2}|^2 = 0.5$. With the laser, this shows as HALF brightness at Port A. With one photon, the student would see Port A fire on approximately half of individual trials - but each trial gives a definite outcome, not a brightness level. The laser brightness is the expectation value of many single-photon measurement outcomes.

Q5 - Quick Reference: The HWP-angle → PBS-output relationship is single-qubit physics. The ability to read brightness as a continuous value is classical.

Extended explanation: This question is designed to build the quantum-classical bridge explicitly. The correct answer has two parts: (1) the state preparation and routing logic (HWP angle determines which port receives light) is identical in the single-photon and many-photon cases; (2) the observable quantity is different - many photons give a continuous brightness, one photon gives a discrete detection event. The law governing both is the same (Born rule / Malus's Law are the same formula at different scales).

Troubleshooting - Lab 1

Symptom	Likely Cause	Fix
Port B never goes dark even at 0° HWP	HWP not at true 0° (relative zero not established), or HWP mounted tilted off normal	Re-establish relative zero: rotate HWP slowly to find the angle of maximum Port A brightness. Mark that as 0°. Check that the HWP mount

		post is vertical and the plate seats flat.
Port A and Port B both show LITTLE regardless of HWP angle	Beam is clipping at the PBS aperture - post height mismatch, or PBS not centered on beam	Use a white card to trace the beam. Adjust post heights so beam enters the PBS at center height. Re-center the PBS laterally.
Superposition step shows MOST/LITTLE instead of HALF/HALF	HWP not at exactly 22.5°, or PBS cube has poor extinction ratio at this wavelength	Confirm the HWP angle scale is read correctly. Try fine-adjusting ±2° around 22.5°. If contrast is poor across all angles, check that the PBS is the 650 nm laser-line version.
Binary message exercise: decoder consistently misreads	Encoder not waiting for beam to stabilize after rotating HWP before signaling decoder	Establish a simple protocol: Encoder says 'ready' after setting the angle; Decoder confirms before recording. The beam settles instantly but human coordination is the common failure.
One port is always brighter than the other at the 22.5° superposition setting	Linear polarizer is not at relative zero, so input polarization is not purely horizontal	Return to Step 1 and re-establish the polarizer's relative zero before proceeding to Step 4.

LAB 2 Superposition with Half-Wave Plate

Goal

- Demonstrate how a half-wave plate alters the polarization of a laser beam and describe the resulting brightness change at each PBS port.
- Explore Malus's Law qualitatively by predicting and observing brightness changes at various HWP angles, then evaluating whether observations are consistent with $I = I_0 \cos^2(\theta)$.

- Explore quantum superposition by observing how polarization state determines photon path through a polarizing beamsplitter.

Theory

Superposition is the idea that an object can exist in two different states at the same time. At the quantum scale, objects can act as both a particle and a wave simultaneously. A classic illustration is Schrödinger's cat: a cat in a sealed box with a 50–50 chance of being poisoned is considered both alive and dead before you open the box. Opening the box - making a measurement - collapses the system to one state.

Malus's Law describes how the brightness of polarized light changes with angle: $I = I_0 \cos^2(\theta)$, where θ is the angle between the light's polarization direction and the polarizer's transmission axis. A half-wave plate rotates linear polarization without absorbing photons. When a HWP is in the beam path and rotated by angle θ, the effective formula becomes $I = I_0 \cos^2(2\theta)$ - the factor of two comes from the double-angle rotation rule of wave plates.

Equipment

- Red laser pointer (650 nm)
- Linear polarizer (always in beam path)
- Half-wave plate (HWP) - zero-order, 650 nm
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position

Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
Half-wave plate (HWP)	Column K, Row 6
Polarizing beamsplitter (PBS)	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

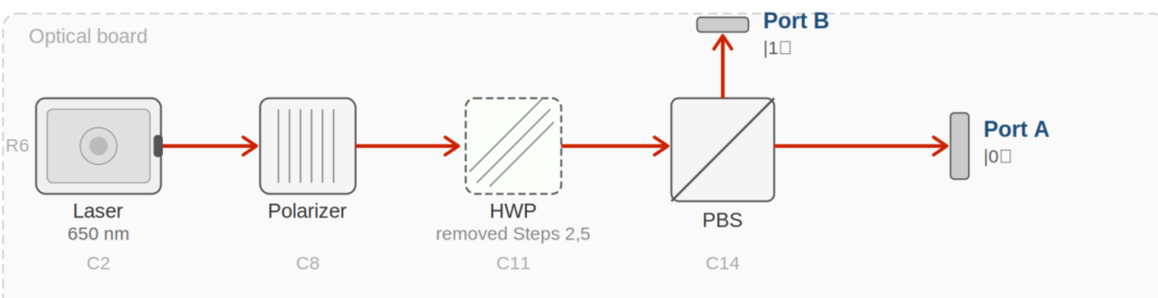

Fig 2.1 — Lab 2 beam path. HWP (dashed border) is removed in Steps 2 and 5.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.

- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ **Safety**

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Place the laser and polarizer. Rotate the polarizer to maximum brightness. This is your ALL.

Brightness (ALL): ____

Step 2 - Add PBS, establish relative zero

Place the PBS at Column N, Row 6. Rotate the polarizer to highest Port A brightness. Record angle as polarizer relative zero.

Polarizer relative zero angle: ____

Port A brightness: ____

Step 3 - Add HWP

Place the HWP at Column K, Row 6. Rotate to maximum Port A brightness. Record as HWP relative zero.

HWP relative zero angle:

____

LQ1) In your own words, what does the HWP do to the polarization of the light?

Answer:

____

____

Step 4 - Extinguish each port in turn

Rotate the HWP until Port B disappears.

LQ2) Why did Port B disappear? What is the new HWP angle relative to zero?

I expect Port B to show ___ because ___.

Answer:

____

____

Port A brightness:

____

LQ3) Is Port A similar to Step 2? Why should it be?

Answer:

____

____

Now rotate the HWP until Port A disappears.

LQ4) Why did Port A disappear and Port B reappear?

Answer:

____

____

Port B brightness:

LQ5) Is this brightness similar to Step 2? If so, why?

Answer:

Step 5 - Qualitative Malus's Law: rotating the polarizer

Remove the HWP. Return polarizer to relative zero. For each angle below, predict then observe.

Prediction guide: 0° → ALL; 45° → HALF; 90° → NONE.

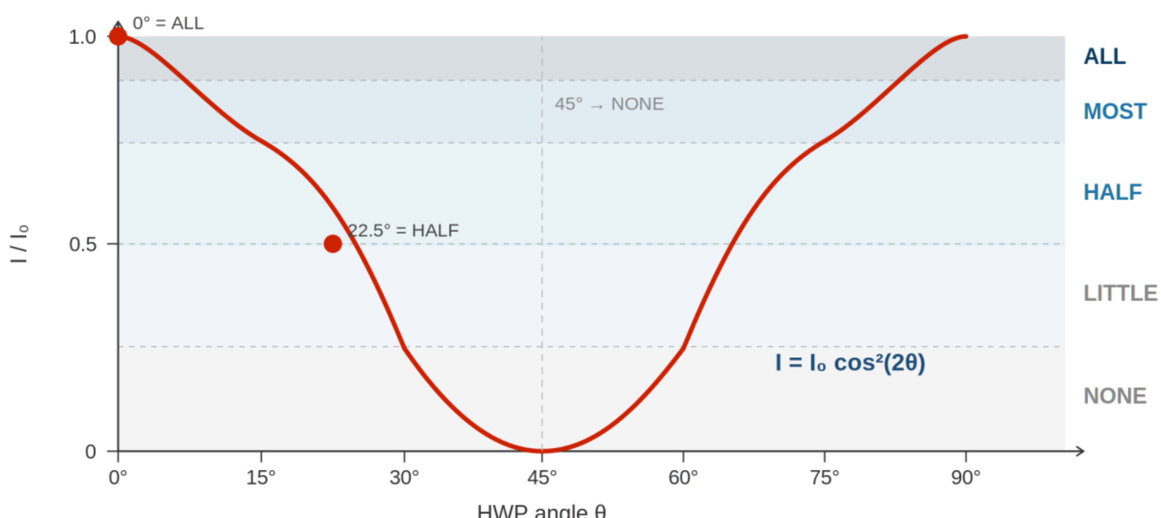

Fig 2.2 — Malus's Law with visual scale overlay. Predict your table entries from this curve before rotating.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Polarizer Angle (°)	Predicted Brightness	Observed Brightness	Does observation match? Write one sentence.
0°			

15°			
30°			
45°			
60°			
90°			

Step 6 - Qualitative Malus's Law: rotating the HWP

Return polarizer to relative zero. Place HWP at relative zero. Rotate the HWP and predict using $I = I_0 \cos^2(2\theta)$.

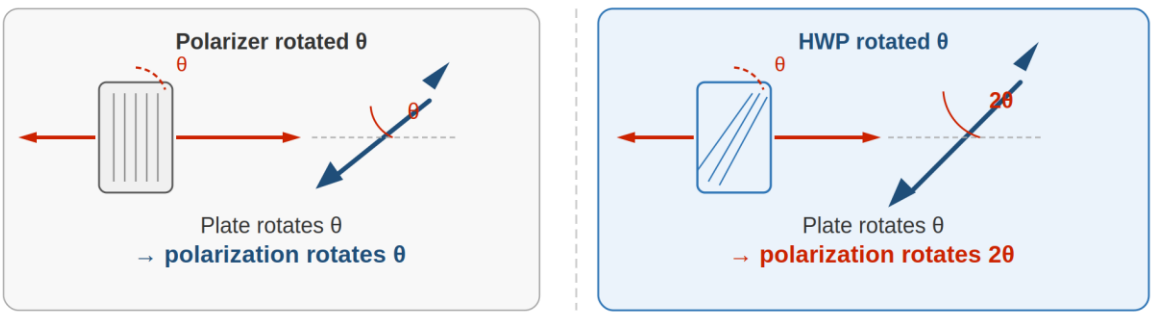

Fig 2.3 — The double-angle rule. A HWP rotates polarization by twice its own plate rotation angle.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

HWP Angle (°)	Predicted Brightness	Observed Brightness	Does observation match? Write one sentence.
0°			
22.5°			
45°			
67.5°			
90°			

Turn off the laser and disassemble.

Post-Lab Questions

6. What is the function of a polarizer, and how does it work? Use an everyday analogy.

____

____

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7. When you rotated the HWP to 45°, Port A went dark. The HWP did not absorb the light - so where did the energy go? What actually caused the brightness at Port A to drop?

   Hint: think about what the PBS does and where both output ports are.

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8. When the HWP was at relative zero, was the light still polarized? How do you know?

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9. In Step 6, why did brightness return to ALL at 90° - the same as at 0°?

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10. What is superposition? Use the language of this lab - ports, HWP angle, brightness - in your answer.

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11. If a single photon moved through the polarizer at 45°, what would happen? How is that different from what you saw with the laser?

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12. Why is this experiment only an approximation of a quantum system?

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INSTRUCTOR PAGE - LAB 2: SUPERPOSITION WITH HALF-WAVE PLATE

NOT FOR STUDENT DISTRIBUTION

Designated Laser Operator (DLO) - Lab 2

Lab 2 DLO: rotate to Student B. If solo, instructor holds DLO role throughout.

Equipment Notes - Lab 2

Equipment note: This lab is the most quantitatively demanding in the visual-observation format. The prediction-then-observe structure for the Malus's Law tables is essential: students must commit to a prediction before rotating anything. Enforce this in the classroom - do not allow students to rotate first and then write a prediction that matches. The intellectual value is in the prediction, not the observation alone.

Equipment note: The factor-of-two difference between the polarizer formula ($\cos^2\theta$) and the HWP formula ($\cos^2(2\theta)$) is a frequent point of confusion. Emphasize: with a polarizer alone, θ is the angle between polarization and the analyzer axis. With an HWP, the plate rotates polarization by 2θ, so the effective angle at the analyzer is 2θ. The formula change reflects a physical difference in what is being rotated.

LQ Answer Key - Lab 2

LQ1 - Quick Reference: The HWP rotates the polarization direction without absorbing or blocking any light.

Extended explanation: This is the key distinction students need between a polarizer (selective, lossy) and a wave plate (rotational, lossless). An ideal HWP transmits 100% of the incident light while changing only the direction of polarization. In practice, there are small reflection and absorption losses, but these are negligible for the visual observation format.

LQ2 - Quick Reference: Port B disappeared because the HWP rotated the polarization to horizontal, sending all light to Port A. The new angle is 0° (or 90° depending on starting orientation) relative to zero.

Extended explanation: When the HWP fast axis aligns with the incoming polarization (or is at 90° to it), the polarization direction is unchanged or rotated 180° - both of which are horizontal for our setup. The PBS then sends all light to Port A and none to Port B.

LQ3 - Quick Reference: Yes, Port A brightness should be similar to Step 2 because the total light in the beam has not changed - only its polarization direction.

Extended explanation: Energy conservation is the key principle here. The HWP does not absorb light; it redirects it. When all light goes to Port A, Port A brightness should equal the original ALL reference (accounting for minor reflection losses in the optics).

LQ4 - Quick Reference: The HWP rotated the polarization to vertical, so the PBS now reflects all light to Port B. Port A is dark because no horizontal component remains.

Extended explanation: This is the NOT gate operation, which students will formalize in Lab 6. Connect explicitly: the 45° HWP setting that extinguishes Port A is the same setting they will use to build the X gate.

LQ5 - Quick Reference: Port B brightness should be similar to the original ALL, for the same energy conservation reason as LQ3.

Extended explanation: If Port B is significantly dimmer than ALL, the most likely cause is a small polarizer misalignment or beam clipping at the PBS. This is a useful self-check step.

Post-Lab Answer Key - Lab 2

Q1 - Quick Reference: A polarizer transmits light oscillating in one specific direction and absorbs or deflects light in all other directions.

Extended explanation: A useful analogy: a picket fence only lets through waves (like jump ropes) aligned with the gaps. Light polarized at 90° to the polarizer axis is completely blocked; light at intermediate angles is partially transmitted according to Malus's Law. Students at the HS/first-year level do not need to know the physical mechanism (birefringent crystal or wire grid), but they should understand the selectivity: a polarizer makes a measurement on the polarization and keeps only one result.

Q2 - Quick Reference: The energy went to Port B - the PBS routed the vertically polarized light there. The HWP did not absorb or destroy the energy.

Extended explanation: This is a correction from earlier editions. The HWP is a unitary (energy-conserving) optical element. The drop in Port A brightness is not caused by absorption in the HWP; it is caused by the PBS routing light to Port B. A careful student might ask: where does Port B light go if no screen is there? It exits the PBS and travels in a different direction - it is still there, just not being observed. This is an important precursor to the measurement problem in quantum mechanics.

Q3 - Quick Reference: Yes, the light is still linearly polarized - the HWP only rotates the direction, it does not change the polarization type.

Extended explanation: At HWP 0°, the fast axis aligns with the incoming polarization. The light passes through with its polarization direction unchanged (or rotated 180°, which is identical for linear polarization). Students can verify this by slowly rotating the analyzer: they will see brightness vary sinusoidally, confirming linear polarization.

Q4 - Quick Reference: At 90°, $\cos^2(2\times90°) = \cos^2(180°) = 1$, same as $\cos^2(0°) = 1$. The polarization has been rotated 180°, which is the same state as 0° for linear polarization.

Extended explanation: This is a key feature of the double-angle rule: the intensity pattern repeats with period 90° for the HWP (vs. 180° for a polarizer alone). Physically, rotating the HWP by 90° produces a 180° polarization rotation - returning to the original direction.

Q5 (superposition) - Quick Reference: Superposition is the state where the qubit is in a combination of $|0\rangle$ and $|1\rangle$ simultaneously, represented optically by polarization at 45° - which gives equal brightness at both PBS ports.

Extended explanation: The best answers will use the lab's own observable: 'When the HWP is at 22.5°, Port A and Port B both show approximately HALF brightness. This means the light is in an equal superposition of horizontal and vertical polarization - neither purely $|0\rangle$ nor purely $|1\rangle$.' Students who answer only abstractly ('superposition means both at once') should be prompted to connect it to what they saw.

Q6 (single photon at 45°) - Quick Reference: The photon would go to either Port A or Port B with 50% probability - a single, definite outcome, not a brightness level.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The new element here: a polarizer at 45° does not 'split' a single photon - it makes a probabilistic measurement. The photon either passes (probability $\cos^2(45°) = 0.5$) or is absorbed/deflected (probability 0.5). There is no partial photon. The brightness students observe with the laser is the average of many such all-or-nothing events.

Q7 (approximation) - Quick Reference: It is an approximation because the laser produces many photons simultaneously, giving continuous average brightness rather than discrete single-photon detection events.

Extended explanation: The experiment is an excellent classical analogue - the physics of polarization rotation and beamsplitting is identical at the single-photon and many-photon levels. What is missing: the randomness of individual outcomes, the impossibility of predicting which port a single photon will take, and the collapse of superposition upon measurement. These are genuinely quantum features that cannot be observed with a laser.

Troubleshooting - Lab 2

Symptom	Likely Cause	Fix
Predicted and observed brightness never match, even at 0° and 90°	Polarizer and HWP zeros not established correctly before the prediction table	Return to Steps 2 and 3. Re-establish both relative zeros with the other element removed. Confirm Port A is ALL at both zeros before proceeding to the table.
HWP Step 6 table shows ALL at 45° instead of HALF	Student is reading the HWP angle scale incorrectly - confusing plate angle with polarization angle	Clarify: the scale on the RSP05 mount reads the plate rotation angle θ. The polarization rotates by 2θ. At plate 45°, polarization rotates 90° - sending all light to Port B, not splitting it. If the student sees ALL at the transmitted port at 45°, check whether they have the HWP and polarizer swapped in position.
Port A shows LITTLE even at HWP relative zero	Beam is partially blocked by aperture or mount hardware between HWP and PBS	Walk the beam with a white card from HWP to PBS. Look for clipping at any mount edge. Adjust post height or lateral position of the intermediate mount.
Brightness does not return to ALL at HWP 90°	HWP 90° is not the true optical zero - the plate may not be reading its own mechanical zero correctly	Verify by slowly sweeping from 0° to 180° and finding two maxima. They should be approximately 90° apart. The midpoint between them is the true optical axis of the system.

LAB 3 Measurement and Basis Selection

Goal

- Demonstrate how polarization basis selection using a half-wave plate encodes binary information into light, forming the foundation of a quantum key distribution (QKD) system.
- Simulate secure quantum communication by comparing sender and receiver basis selections and identifying which shared measurements form a valid cryptographic key.
- Illustrate how eavesdropping on a quantum channel is detectable: observe how an interceptor's random basis choices reduce the sender-receiver match rate.

Theory

In quantum protocols, measurement is not passive: observing a qubit in one basis disturbs what would have been true in another. That is why polarization-based schemes can reveal eavesdropping. An interceptor who measures in a random basis and re-sends light injects extra randomness into the bits the receiver records.

Rotating a half-wave plate changes which linear basis the light is 'written in' before reaching the PBS. When both parties choose random bases, about half of all trials match. If an eavesdropper measures in a random basis and re-sends, the fraction of agreeing trials drops toward about one quarter. That statistical change is how eavesdropping gets detected.

Equipment

- Red laser pointer (650 nm)
- Linear polarizer (always in beam path)
- Half-wave plate (HWP) - zero-order, 650 nm
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
Half-wave plate (HWP)	Column K, Row 6
Polarizing beamsplitter (PBS)	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

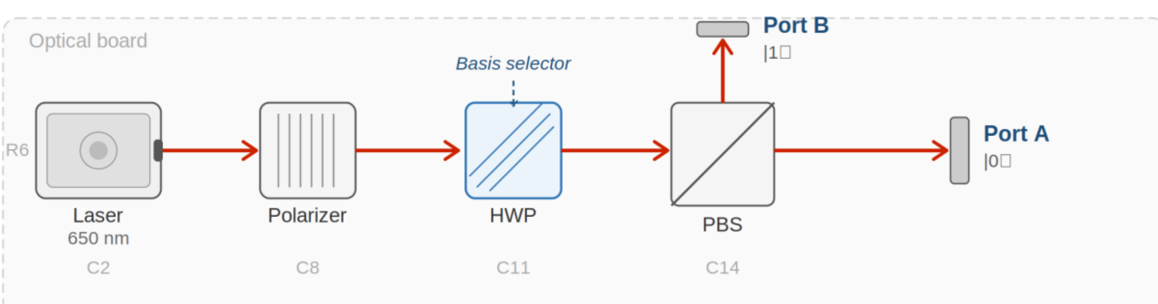

Fig 3.1 — Lab 3 beam path. The HWP (highlighted) is the basis selector for the QKD simulation.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Laser and polarizer only. Rotate to maximum brightness. This is ALL.

Step 2 - Establish sender basis table

HWP at Column K, Row 6 at 0°. PBS at Column N, Row 6.

LQ1) What is the role of the linear polarizer in this lab?

Answer:

____

____

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Sender key: + basis 0° = bit 0; + basis 90° = bit 1; × basis 45° = bit 1; × basis −45° = bit 0

Basis	HWP Angle	Bit Sent
+ basis	0°	0
+ basis	90°	1
× basis	45°	1
× basis	−45°	0

LQ2) Could the receiver tell which basis the sender used just by watching which port lit up?

I expect ___ because ___.

Answer:

____

____

____

Step 3 - Sender/receiver comparison (no eavesdropper)

Receiver key: + basis = 0°; × basis = 45°

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Sender Angle	Sender Bit	Receiver Basis	Port A lit?	Port B lit?	Same Basis?
0°	0	+ (0°)			
90°	1	+ (0°)			
45°	1	× (45°)			
−45°	0	× (45°)			
0°	0	× (45°)			
90°	1	× (45°)			
45°	1	+ (0°)			
−45°	0	+ (0°)			

Step 4 - Eavesdropper simulation

Simulate an eavesdropper who intercepts the beam, measures in a random basis, then re-sends.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Eavesdropper Angle	Eaves. Bit	Receiver Basis	Port A lit?	Port B lit?	Same Basis?

LQ3) When would the receiver not detect the eavesdropper? What is the approximate chance of that?

Compare match rates: no eavesdropper vs. with one. I expect the rate to change from ___ to ___ because ___.

Answer:

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Turn off the laser and put the apparatus away.

Post-Lab Questions

13. Why is superposition such a valuable property for secure cryptography?

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14. What result does the receiver get with the same basis as the sender? With the opposite basis?

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15. Why must basis selection be random?

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16. Name two conditions required for quantum communication to be secure against eavesdropping.

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17. We used a laser rather than single photons. Does that affect the validity of the basis-selection principle? Why or why not?

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INSTRUCTOR PAGE - LAB 3: MEASUREMENT AND BASIS SELECTION

NOT FOR STUDENT DISTRIBUTION

Designated Laser Operator (DLO) - Lab 3

Lab 3 DLO: rotate back to Student A. Remind both students that the Encoder role in the message exercise is not the DLO role - the DLO is responsible for laser safety only, not for the communication game.

Equipment Notes - Lab 3

Equipment note: Lab 3 is conceptually the most direct quantum cryptography demonstration in the sequence. The key physical insight is that the × basis (HWP at ±45°) produces a 50/50 split at a PBS aligned in the + basis - so the receiver using the wrong basis gets a random result even with a perfect channel. This is not a noise effect; it is a fundamental consequence of incompatible bases. Emphasize this distinction when reviewing the eavesdropper table results.

Equipment note: With one or two students per kit, the eavesdropper simulation may require the same student to play multiple roles sequentially rather than simultaneously. This is fine - the conceptual lesson is preserved. Have the student record the 'eavesdropper' measurement, then physically reset the HWP to simulate re-sending, and record the receiver result separately.

LQ Answer Key - Lab 3

LQ1 - Quick Reference: The linear polarizer establishes a known, fixed input polarization so that the HWP rotations produce predictable, reproducible basis states.

Extended explanation: Without the polarizer, the laser's output polarization may not be stable or well-defined (many laser pointers are unpolarized or partially polarized). The polarizer ensures the system starts from a defined $|0\rangle$ or $|1\rangle$ input before the HWP applies any basis rotation. This is the optical analogue of 'state preparation' in a quantum circuit.

LQ2 - Quick Reference: No - when the sender uses the × basis (HWP at ±45°), the PBS shows a 50/50 split. The receiver cannot determine whether the sender set 45° or −45°, so they cannot identify the basis.

Extended explanation: This is the heart of BB84 protocol security. The receiver sees either a definite port (same basis) or a random 50/50 result (different basis) - but they cannot distinguish 'random because wrong basis' from 'random by chance' without communicating with the sender over a classical channel. The security comes from this indistinguishability.

LQ3 - Quick Reference: If the eavesdropper guesses the same basis as both sender and receiver, the interception goes undetected. The probability is approximately 25% per trial (50% chance eavesdropper matches sender × 50% chance eavesdropper matches receiver).

Extended explanation: This is the statistical foundation of BB84 security analysis. With 8 trials, the probability that the eavesdropper goes completely undetected is approximately $0.25^8 \approx 0.00015$ - essentially zero for a large key. The security scales exponentially with key length, which is why even simple QKD protocols become very secure with enough trials. Note: the exact probability depends on the protocol details; this simplified analysis assumes independent random basis choices.

Post-Lab Answer Key - Lab 3

Q1 - Quick Reference: Superposition means a qubit measured in the wrong basis gives a random result, not a reproducible one. An eavesdropper cannot copy an unknown quantum state without disturbing it.

Extended explanation: The no-cloning theorem (which students do not need to know by name) underlies this: an eavesdropper cannot make a perfect copy of a qubit in an unknown state. Any measurement they make collapses the superposition, and their re-sent signal will have a different statistical distribution than the original. This shows up as a reduced match rate in the receiver's results - the statistical signature of interception.

Q2 - Quick Reference: Same basis: receiver gets a definite result matching the sender's bit. Different basis: receiver gets a random result (approximately 50/50).

Extended explanation: This is the operational definition of basis compatibility in quantum measurement. When bases match, the receiver's PBS is aligned to the sender's encoding axis - the measurement is deterministic. When they differ by 45°, the light is in superposition relative to the receiver's basis, and the outcome is probabilistic. This is not an imperfection; it is a fundamental feature of quantum measurement.

Q3 - Quick Reference: If basis selection were not random, an eavesdropper could predict the basis and always intercept without detection.

Extended explanation: Randomness is the source of security in QKD. If the sender always used the + basis, the eavesdropper would simply always measure in the + basis and pass through undetected. Random basis selection means the eavesdropper cannot know which basis to use, and any wrong-basis measurement introduces detectable errors.

Q4 - Quick Reference: Any two of: (1) the quantum channel must be authenticated so the receiver knows they are talking to the real sender; (2) basis reconciliation must occur over a classical channel; (3) error rate in the reconciled key must be below a threshold; (4) privacy amplification must be applied to remove any information an eavesdropper may have gained.

Extended explanation: Students are not expected to know all four in detail - the question is designed to prompt them to think beyond the optical bench. A good answer names at least two and explains why they matter. The most important for this lab's context are authentication (to prevent man-in-the-middle attacks) and error rate checking (to detect interception).

Q5 - Quick Reference: Using a laser rather than single photons does not affect the basis-selection principle - the same angular relationships between sender, receiver, and eavesdropper bases apply.

Extended explanation: The statistical argument (match rate drops from ~50% to ~25% with eavesdropping) holds for the laser demonstration because we are observing the same angular relationships. What is missing with a laser: the irreversibility of single-photon measurement. With a laser, a beam splitter can sample the beam without fully destroying it - a real eavesdropper with quantum equipment faces no such luxury with single photons.

Troubleshooting - Lab 3

Symptom	Likely Cause	Fix
× basis angles (±45°) show ALL at one port instead of HALF/HALF	HWP is not at a true ±45° relative to the polarizer's zero - the relative zero was not established correctly	Re-establish polarizer relative zero first (Step 2, Port A = ALL). Then set HWP to the angle that gives HALF/HALF. If that angle is not near ±45°, the mechanical zero of the mount does not match the optical zero - note the offset and use it for all subsequent angle settings.
Eavesdropper simulation shows same match rate as no-eavesdropper case	The eavesdropper is inadvertently using the same basis as both sender and receiver every trial	Randomize the eavesdropper basis choices explicitly - write them on slips of paper drawn blind, rather than letting the student choose. This also makes the statistical argument more compelling.
Students cannot distinguish 'same basis' from 'different basis' outcomes in the table	Students are not clear on when the PBS gives a definite vs. random result	Run a focused demonstration: sender at 0° (+ basis), receiver PBS in + configuration - show ALL at Port A. Then rotate sender HWP to 45° (× basis, same bit) without telling the receiver which basis - show 50/50. Ask: can the receiver tell which basis was used? Answer: no. This makes the indistinguishability concrete.

LAB 4 Quarter-Wave Plate and Circular Polarization

Goal

- Demonstrate how a quarter-wave plate at 45° converts linearly polarized light into circularly polarized light and observe the resulting approximately equal brightness split at the PBS ports.
- Illustrate measurement basis incompatibility: show that a linear PBS cannot distinguish circular polarization states without first converting them back using a second QWP.
- Connect circularly polarized light at a beamsplitter to the quantum principle that measuring a qubit in an incompatible basis yields uncertain rather than definite outcomes.

Theory

A quarter-wave plate (QWP) is a birefringent element with a fast axis and a slow axis. Light along the fast axis travels slightly faster than light along the slow axis, building up a quarter-wavelength phase difference between the two components. When linearly polarized light enters at 45° to those axes, that delay converts straight-line oscillation into circular rotation - the light is now circularly polarized.

The PBS separates light by linear orientation: horizontal to Port A, vertical to Port B. Circularly polarized light contains equal amounts of both orientations, rotating continuously, so the PBS cannot favor one port. You will see an approximately equal split. No adjustment of the PBS alone can distinguish right-hand from left-hand circular polarization. To recover that information you need a second QWP to convert the light back to linear before the PBS - that is basis conversion in hardware.

This is the direct optical analogue of measuring a qubit in the wrong basis: the result appears random, not because the information is gone, but because the detector cannot access it in that form.

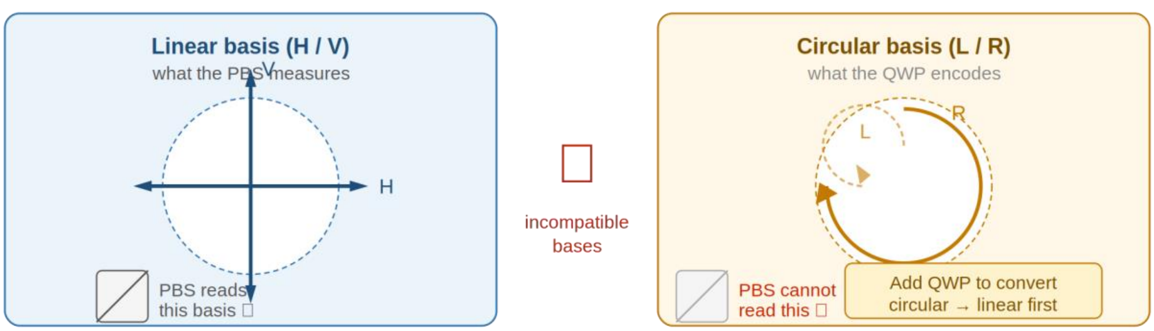

Fig 4.2 — Basis incompatibility. The PBS measures H/V only. Circular polarization requires a QWP to convert before the PBS can read it.

Equipment

- Red laser pointer (650 nm)
- Linear polarizer (always in beam path)

- Quarter-wave plate (QWP) - zero-order, 650 nm - two needed
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
QWP, first	Column K, Row 6
QWP, second (added in Step 5)	Column L, Row 6
Polarizing beamsplitter (PBS)	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

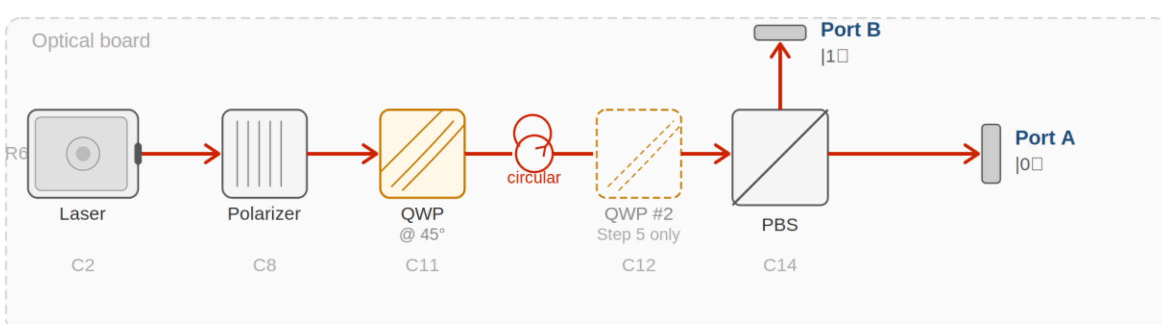

Fig 4.1 — Lab 4 beam path. QWP at 45° converts linear to circular polarization (circle symbol). QWP #2 (dashed) added in Step 5.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Laser and polarizer only. Rotate to maximum brightness. This is ALL.

Step 2 - Linear baseline

QWP at Column K, Row 6 at 0°. PBS at Column N, Row 6. Light remains linear.

Port A brightness: ____

Port B brightness: ____

Step 3 - Create circular polarization

Rotate the QWP to 45°.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Observed Brightness
Port A (|0⟩ , transmitted)	
Port B (|1⟩ , reflected)	

LQ1) Is the split close to equal? What does that tell you about how the PBS 'sees' circularly polarized light?

I expected ___ because ___. What I actually saw was ___.

Answer: ____
____

Step 4 - Try to distinguish circular from linear by rotating the PBS

With QWP at 45°, try adjusting the PBS orientation slightly. Does the split change significantly?

What happens to the Port A / Port B split when you adjust:

____

____

LQ2) Why does the equal split persist? What does this tell you about a linear detector's ability to read circular polarization?

Answer:

____

____

____

Step 5 - Add second QWP to recover the basis

Place second QWP at Column L, Row 6 at 45°.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Setup	Port A Brightness	Port B Brightness	Equal or One-Sided?
QWP at 45°, no second QWP			
Second QWP at +45°			
Second QWP at −45°			

LQ3) After adding the second QWP at 45°, what changed? Why did the equal split disappear?

I expect ___ because the second QWP ___.

Answer:

____

____

____

LQ4) After rotating the second QWP to −45°, where did most of the light go? What does that tell you about right-hand vs left-hand circular polarization?

Answer:

____

____

____

Turn off the laser and put apparatus away.

Post-Lab Questions

18. What is the job of a quarter-wave plate at 45° when the incoming light is linearly polarized?

    ____

    ____

    ____

    ____

19. Why does circularly polarized light produce an approximately equal split at a PBS?

    ____

    ____

    ____

    ____

20. What does it mean to say the PBS measures in the H/V basis? Why can it not distinguish right from left circular without extra wave plates?

    ____

    ____

    ____

    ____

21. In one sentence: what is measurement-basis mismatch in this lab, and how is it like measuring a qubit in the wrong basis?

22. If a single photon were in a circular polarization state and hit the PBS, what would happen at each port?

    Hint: HALF brightness with a laser means half the photons go each way on average. With one photon, what does that mean?

23. We used a laser. What is approximate about calling this a quantum measurement demonstration, and what would stay the same with one photon at a time?

INSTRUCTOR PAGE - LAB 4: QUARTER-WAVE PLATE AND CIRCULAR POLARIZATION

NOT FOR STUDENT DISTRIBUTION

Designated Laser Operator (DLO) - Lab 4

Lab 4 DLO: rotate to Student B. Note that Lab 4 uses two QWPs simultaneously in Step 5 - confirm both mounts are available and correctly positioned before energizing the laser.

Equipment Notes - Lab 4

Equipment note: Lab 4 introduces the QWP, which behaves very differently from the HWP students have used so far. The key distinction to reinforce: a QWP introduces a quarter-wavelength (π/2 radian) phase delay between the fast and slow axis components, compared to the HWP's half-wavelength (π radian) delay. At 45° to the input polarization, the QWP converts linear to circular; the HWP at 45° converts linear to linear (orthogonal). Students sometimes conflate the two - the observable difference (50/50 at any PBS orientation for circular vs. deterministic routing for linear) is the best way to make this concrete.

Equipment note: Zero-order QWPs at 650 nm are particularly sensitive to angular tilt. When the QWP does not sit perpendicular to the beam, the retardation deviates from exactly λ/4, producing elliptical rather than circular polarization. Students will observe a split that is approximately but not exactly 50/50 - this is physically correct and should be framed as 'approximately circular' rather than an error. The second QWP step (Step 5) is a more sensitive test of circularity: a perfect circular input will give a definite output after the second QWP; an elliptical input will give a slightly impure result.

LQ Answer Key - Lab 4

LQ1 - Quick Reference: The split is approximately equal (HALF/HALF). The PBS 'sees' equal horizontal and vertical components because circularly polarized light has equal projections onto all linear axes.

Extended explanation: Circularly polarized light can be decomposed as $(1/\sqrt{2})(|H\rangle + i|V\rangle)$ for right-circular or $(1/\sqrt{2})(|H\rangle - i|V\rangle)$ for left-circular. The magnitude of the H and V components is equal in both cases: $|1/\sqrt{2}|^2 = 0.5$. The PBS measures in the H/V basis and finds a 50% probability for each. The phase factor i carries the information about handedness - but the PBS cannot read phase, only amplitude. Hence the 50/50 split for both circular states.

LQ2 - Quick Reference: The 50/50 split persists because circularly polarized light has equal projections onto all linear polarization axes - there is no orientation of the PBS that favors one port.

Extended explanation: This is the physical meaning of 'basis incompatibility.' A measurement device that works in one basis (linear H/V) cannot extract information encoded in a different basis (circular L/R) without a basis conversion element. Students sometimes expect that rotating the PBS will help - it does not, because all linear bases are equally mismatched to circular polarization.

LQ3 - Quick Reference: Adding the second QWP converts the circular polarization back to linear, restoring a definite PBS output - one port goes to ALL or MOST, the other to NONE or LITTLE.

Extended explanation: The second QWP undoes what the first QWP did - it is the optical basis conversion from circular back to linear. Mathematically: QWP(+45°) followed by QWP(+45°) = HWP at 45°, which rotates polarization 90°. The PBS can now read the converted linear state deterministically. This is the hardware implementation of a basis change in quantum measurement.

LQ4 - Quick Reference: With the second QWP at −45°, most of the light goes to the opposite port compared to +45°. The QWP distinguishes right-hand from left-hand circular polarization by converting each to a different linear state.

Extended explanation: At +45°, right-circular converts to horizontal (→ Port A) and left-circular converts to vertical (→ Port B). At −45°, the conversion is reversed. This is the only way to distinguish the two circular states - a second QWP oriented at ±45° acts as a circular polarization analyzer. Without it, the PBS is blind to handedness.

Post-Lab Answer Key - Lab 4

Q1 - Quick Reference: A QWP at 45° introduces a quarter-wavelength phase delay between the components along its fast and slow axes, converting linear polarization into circular polarization.

Extended explanation: More precisely: linear polarization at 45° to the QWP axes has equal amplitude along the fast and slow axes. The QWP adds a 90° (π/2 radian) phase delay to the slow component. Equal amplitudes with a 90° phase difference is the definition of circular polarization. At other input angles, the output is elliptically polarized (not purely circular). The 45° input angle is the special case that produces perfect circularity with a zero-order QWP.

Q2 - Quick Reference: Circular polarization has equal amplitude components along any pair of orthogonal linear axes, so the PBS - which separates H from V - finds equal amounts to route to each port.

Extended explanation: This is worth connecting to the quantum measurement formalism for instructors comfortable with that language. The circular state $|R\rangle = (1/\sqrt{2})(|H\rangle + i|V\rangle)$ has probability amplitudes of $1/\sqrt{2}$ for both H and V. The Born rule gives P(H) = P(V) = 1/2. The PBS is a projective measurement in the $\{|H\rangle, |V\rangle\}$ basis, and the probabilities are equal regardless of the phase between H and V - which is why rotating the PBS does not help.

Q3 (PBS H/V basis) - Quick Reference: The PBS measures in the H/V basis: it routes horizontal light one way and vertical light another. It cannot distinguish circular states because both R and L circular polarization have equal H and V content - the difference between them is a phase, not an amplitude, and the PBS is insensitive to phase.

Extended explanation: This is perhaps the deepest conceptual point in Lab 4. Phase information is invisible to a projective measurement in an incompatible basis. A QWP is needed to convert phase differences into amplitude differences before the PBS can read them. This is directly analogous to the quantum computing principle that information encoded in a superposition cannot be accessed by a measurement in an incompatible basis - you always need to apply the right gate first.

Q4 (basis mismatch in one sentence) - Quick Reference: Measurement-basis mismatch means the detector (PBS in H/V basis) cannot access information encoded in the circular basis - just as measuring a qubit in the Z basis gives a random result when the qubit is in the X-basis eigenstate.

Extended explanation: Students may initially answer this vaguely. A good answer names the two bases explicitly (linear H/V and circular L/R), states what the PBS finds (50/50, random), and connects it to quantum measurement (measuring in the wrong basis destroys the encoded information).

Q5 (single photon, circular) - Quick Reference: A single photon in a circular state would be detected at Port A or Port B with 50% probability each - a single, definite, random outcome. The laser's HALF brightness is the average of many such outcomes.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The specific addition here: the handedness of the circular polarization is preserved in the quantum state, but a single PBS measurement in the H/V basis cannot read it. To read handedness, a second QWP must be applied before the measurement - just as in Step 5 of the procedure.

Troubleshooting - Lab 4

Symptom	Likely Cause	Fix
QWP at 45° does not produce a 50/50 split - one port is much brighter	QWP is tilted off normal (most common with zero-order plates), or the angle is not truly 45° relative to the input polarization	Confirm the QWP mount post is vertical and the plate is seated flat. Re-establish the QWP's 45° angle relative to the polarizer's zero. Check by rotating the QWP slowly - the split should be most equal near 45°.
Second QWP does not restore a definite PBS output in Step 5	First QWP is producing elliptical rather than circular polarization (tilt or angle error), so the second QWP cannot fully convert it back to linear	Remove the second QWP. Check the first QWP angle and tilt. A rough test: if rotating the first QWP slightly from 45° makes the split more unequal in both directions, it is close to optimal. If one direction makes it more equal, the current angle is off.
Steps 4 and 5 look identical - second QWP seems to make no difference	Second QWP is at 0° (not 45°) - it is not converting circular to linear	Confirm the second QWP is set to 45° relative to the polarizer zero, not relative to its own mechanical zero. The angle reference for all wave plates in this lab is the polarizer's zero.
Students observe a 50/50 split even with QWP at 0°	Input polarization is not purely horizontal - polarizer is not at its relative zero	Return to Step 1 and re-establish the ALL reference with the polarizer at true relative zero. Any residual tilt in the polarizer mount produces a mixed input state.

LAB 5 Hadamard Gate Implementation

Goal

- Demonstrate how a half-wave plate at exactly 22.5° implements the quantum Hadamard gate by transforming a definite polarization state into an equal superposition of |0⟩ and |1⟩ .
- Predict the output brightness at each PBS port before setting the HWP angle, then compare predictions to observations.
- Verify that the Hadamard gate is its own inverse by applying it twice and confirming the light returns to its original polarization state.

Theory

The Hadamard gate is one of the most important tools in quantum computing. It takes a qubit that is definitely in one state - say |0⟩ - and puts it into an equal superposition of |0⟩ and |1⟩ . After the gate, the qubit is genuinely both states at once until a measurement forces a choice. Apply H twice, and you are back where you started: H × H = I.

In this lab, the Hadamard gate is a half-wave plate at exactly 22.5°. Why that angle? A HWP rotates polarization by twice its own rotation angle. A 22.5° plate rotation gives a 45° polarization rotation - splitting the beam equally between horizontal (|0⟩) and vertical (|1⟩) at the PBS. No fancy electronics: just one plate at one angle. That is the lesson.

The Hadamard gate as an arrow diagram

Read the arrows below. No arithmetic needed - just understand what the gate does to each input state.

Input state	→	Output at Port A (|0⟩)	Output at Port B (|1⟩)
|0⟩ (horizontal)	→	HALF	HALF
|1⟩ (vertical)	→	HALF	HALF

Both inputs give the same result: equal brightness at both ports. That equal split is the Hadamard gate's signature - it erases the difference between |0⟩ and |1⟩ and replaces it with genuine uncertainty. Apply H twice, and the second gate undoes the first exactly: H × H = I.

Equipment

- Red laser pointer (650 nm)
- Linear polarizer (always in beam path)
- Half-wave plate (HWP) - zero-order, 650 nm - two needed for H×H step
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
HWP, first	Column K, Row 6
HWP, second (added in Step 5)	Column L, Row 6
PBS	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

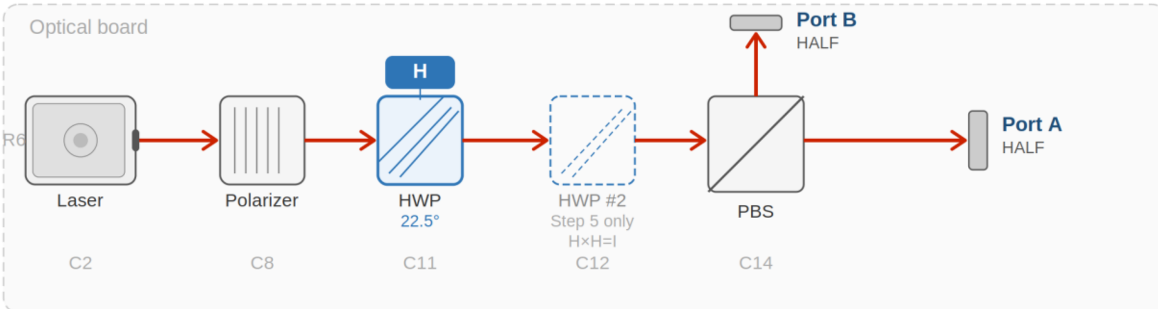

Fig 5.1 — Lab 5 beam path. HWP at 22.5° = Hadamard gate H (blue). HWP #2 (dashed) verifies H × H = I in Step 5.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.

- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Laser and polarizer only. Maximum brightness. This is ALL.

Step 2 - Baseline: input state |0⟩

HWP at C11 at 0°. PBS at C14. Input is horizontal (|0⟩). Predict before observing.

Predicted Port A:

____

____

Predicted Port B:

____

____

Observed Port A:

____

____

Observed Port B:

____

____

LQ1) Does the observation match the prediction from the arrow diagram? Does this make sense for a definite input state?

Answer:

____

____

____

Step 3 - Apply the Hadamard gate (HWP to 22.5°)

Rotate the HWP to exactly 22.5°. Predict first using the arrow diagram.

Predicted Port A:

____

____

Predicted Port B:

____

____

My reasoning:

____

____

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Predicted	Observed	Match? Write one sentence.
Port A (|0⟩)			
Port B (|1⟩)			

LQ2) Why does 22.5° specifically produce this equal split - and not 10° or 30°?

A HWP rotates polarization by twice its own angle. At 22.5°, polarization rotates by ___°, which means ___.

Answer:

____

____

____

Step 4 - Input state |1⟩

Rotate polarizer 90° to create |1⟩ input. Keep HWP at 22.5°. Predict first.

Predicted Port A:

____

____

Predicted Port B:

____

____

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Predicted	Observed	Match?
Port A (|0⟩)			
Port B (|1⟩)			

LQ3) Both |0⟩ and |1⟩ inputs gave HALF/HALF. What does the Hadamard gate do to information about the original state?

Answer:

____

____

____

Step 5 - Verify H × H = I

Return polarizer to horizontal. Add second HWP at C12 at 22.5°. Predict first.

Predicted Port A:

____

____

Predicted Port B:

____

____

My reasoning:

____

____

▼ **Visual Observation Scale** ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Predicted	Observed
Port A (|0⟩)		
Port B (|1⟩)		

LQ4) Did the two HWPs cancel and return the light to |0⟩ ? What property of the Hadamard gate does this demonstrate?

H × H = I means ___. In this lab that showed up as ___.

Answer:

____

____

Turn off the laser and put the apparatus away.

Post-Lab Questions

24. What is the Hadamard gate, and what does it do to a definite input state like $|0\rangle$ or $|1\rangle$?

    ____

    ____

25. Why does the HWP need to be at exactly 22.5°? What goes wrong at 10° or 45°?

    ____

    ____

26. What does the arrow diagram show? How does it help you predict before rotating anything?

    ____

    ____

27. You showed H × H = I. Why is being its own inverse useful in quantum algorithm design?

28. If a single photon in $|0\rangle$ passed through an HWP at 22.5° and hit the PBS, what would happen?

HALF brightness with a laser means half the photons go each way on average. With one photon, what does that mean?

29. We used a laser. In one sentence, why is this still a useful model for a single-qubit Hadamard gate, and what would look different one photon at a time?

INSTRUCTOR PAGE - LAB 5: HADAMARD GATE IMPLEMENTATION

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Designated Laser Operator (DLO) - Lab 5

Lab 5 DLO: rotate to Student A. Lab 5 is the most conceptually dense lab in the sequence - the DLO should be the student who has shown stronger engagement with the quantum formalism so far. Instructor should circulate frequently during the predict-before-observe steps.

Equipment Notes - Lab 5

Equipment note: Lab 5 uses an HWP, not a QWP - this distinguishes it fundamentally from Lab 4. Both labs show a 50/50 PBS split, but for entirely different physical reasons: Lab 4 produces circular polarization (phase relationship between H and V); Lab 5 produces diagonal linear polarization (equal H and V amplitudes, no phase relationship). A PBS cannot tell these apart - they both show HALF/HALF. This is worth making explicit: two physically different states can look identical to a given measurement device. This is the essence of basis incompatibility.

Equipment note: The predict-before-observe requirement in this lab is pedagogically critical. The Hadamard arrow diagram is designed to allow students to predict without calculation - they just read the table. If students are filling in predictions after rotating the HWP, they are not engaging with the diagram's purpose. Enforce prediction-first as a classroom norm before starting Lab 5.

Equipment note: The H × H = I step (Step 5) is the most satisfying demonstration in the lab. If students have correctly predicted HALF/HALF for Step 3 and then see ALL at Port A after adding the second HWP, the conceptual impact is strong. Give students a moment to sit with that result before explaining it.

LQ Answer Key - Lab 5

LQ1 - Quick Reference: At 0° HWP, Port A shows ALL and Port B shows NONE - a definite |0⟩ output for a definite |0⟩ input.

Extended explanation: This baseline step is important because it confirms the system is working before the Hadamard gate is applied. A definite input in a definite basis gives a definite output - no superposition, no probability. If students see anything other than ALL/NONE at 0° HWP, there is an alignment problem to address before proceeding.

LQ2 - Quick Reference: 22.5° produces the equal split because the double-angle rule gives 22.5° × 2 = 45° polarization rotation. At 45° from horizontal, the light has equal H and V components: $\cos^2(45°) = \sin^2(45°) = 0.5$.

Extended explanation: This is the mathematical definition of the Hadamard gate in optical terms. At 10°, the polarization rotates only 20° - strongly weighted toward H, giving MOST at Port A and LITTLE at Port B. At 30°, the rotation is 60° - closer to equal but still weighted. Only at 22.5° (rotation to 45°) is the split exactly equal. Students should understand that 22.5° is not arbitrary - it is the unique angle that produces the equal superposition that defines H.

LQ3 - Quick Reference: The Hadamard gate erases information about the input state: both |0⟩ and |1⟩ produce the same HALF/HALF output. After applying H, you cannot determine from the PBS output alone whether the input was |0⟩ or |1⟩.

Extended explanation: This is a deep point. The Hadamard gate does not destroy information - it encodes it in the phase relationship between H and V, which the PBS cannot read. Applying H again recovers the information (H × H = I). The information has been moved into a degree of freedom (phase) that is invisible to the H/V measurement basis. This is directly analogous to how quantum algorithms temporarily hide information in superposition states during computation.

LQ4 - Quick Reference: Yes - two HWPs at 22.5° cancel out, returning light to ALL at Port A. This demonstrates that H is its own inverse: H × H = I.

Extended explanation: The physical explanation: the first HWP rotates polarization +45°; the second rotates it another +45°, giving a total 90° rotation from horizontal - but wait, that should give vertical (|1⟩), not horizontal. The resolution: the second HWP acts on the polarization direction left by the first HWP. The first HWP leaves diagonal (45°) polarization. The second HWP at 22.5° rotates that diagonal polarization back to horizontal. The net effect of two HWPs at 22.5° is a 0° net rotation - identity. The Jones matrix calculation confirms: [H][H] = I (identity matrix).

Post-Lab Answer Key - Lab 5

Q1 - Quick Reference: The Hadamard gate takes a definite input state (|0⟩ or |1⟩) and produces an equal superposition of |0⟩ and |1⟩ , shown optically as a HALF/HALF brightness split at the PBS.

Extended explanation: The Hadamard gate matrix is H = (1/√2)[[1,1],[1,−1]]. The arrow diagram in the student lab shows the magnitude of this: both input states give equal output probabilities. The minus sign in the (1,1) entry carries phase information about the |1⟩ input that is invisible to a simple brightness measurement but would be revealed if a second H gate were applied. This is the level of detail appropriate for the instructor - students need only the arrow diagram concept.

Q2 - Quick Reference: At 10°, the polarization rotates only 20° - giving a strongly unequal split (MOST at Port A, LITTLE at Port B). At 45°, the polarization rotates 90° - giving ALL at Port B and NONE at Port A. Only 22.5° gives the equal split defining the Hadamard gate.

Extended explanation: This answer reinforces the double-angle rule and the uniqueness of 22.5°. A useful exercise: have students predict the Port A brightness at 10°, 22.5°, 30°, and 45° using $\cos^2(2\theta)$ before confirming with the HWP. This connects Lab 5 back to the formula introduced in Lab 7 (or foreshadows it if done in sequence order).

Q3 (arrow diagram) - Quick Reference: The arrow diagram shows what the Hadamard gate does to each input: both |0⟩ and |1⟩ produce equal output at both ports. It allows prediction without calculation by reading the output brightness level directly from the table.

Extended explanation: The arrow diagram is a simplified Jones calculus output - the magnitudes (but not the phases) of the transformation. For HS/first-year students, this is the appropriate level of abstraction. Instructors comfortable with linear algebra can point out that the full matrix also shows the relative phases between H and V components, which carry information that cannot be read by a simple brightness measurement.

Q4 (H × H = I usefulness) - Quick Reference: H × H = I means applying H twice returns the qubit to its original state. This allows quantum algorithms to 'open' a superposition, perform operations, and 'close' it back - using H as a reversible transformation.

Extended explanation: Concrete examples accessible at this level: the Deutsch algorithm uses H to prepare superposition, applies a function, then uses H again to extract a global property of the function in a single query. Grover's algorithm uses H to spread amplitude across all states, then iteratively amplifies the target state. In both cases, H is applied and then its inverse (which is H itself) is applied - the self-inverse property is not a curiosity but a design requirement.

Q5 (single photon through H gate) - Quick Reference: A single photon in $|0\rangle$ would be detected at Port A or Port B with 50% probability each - a single, definite, random outcome on each trial.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The specific addition: the Hadamard gate genuinely creates quantum uncertainty for a single photon - there is no hidden variable that determines in advance which port fires. The laser's HALF brightness is the average of many such fundamentally random events. This is the point at which quantum randomness (intrinsic) differs from classical randomness (ignorance of a hidden variable). At the HS/first-year level, stating this distinction clearly is sufficient without requiring a proof.

Troubleshooting - Lab 5

Symptom	Likely Cause	Fix
H × H step (two HWPs at 22.5°) does not return to ALL at Port A	The two HWPs are not both at exactly 22.5°, or they are oriented relative to different reference zeros	Both HWPs must be set to 22.5° relative to the same reference zero - the polarizer's horizontal direction. Check each HWP individually (with the other removed): each alone should give HALF/HALF at 22.5°. If both pass this test but the combined result is wrong, the mounts may have slightly different scale offsets - fine-tune one HWP while monitoring Port A.
Students predict wrong values from the arrow diagram before observing	Students are misreading the diagram - treating the columns as sequential steps rather than input/output states	Walk through the diagram explicitly before Step 3: point to the $|0\rangle$ row, trace the arrow to the output column, read HALF at Port A and HALF at Port B. Ask: what does HALF mean

		physically? (Equal brightness.) What does that look like on the screen? Then let them observe.
At 22.5°, one port shows MOST and the other LITTLE instead of HALF/HALF	Input polarization is not purely horizontal - polarizer not at relative zero	The Hadamard gate is sensitive to input state. If the input is not $\|0\rangle$ but some other linear state, the output split will not be exactly 50/50. Return to Step 1 and re-establish the polarizer zero.
$\|1\rangle$ input step (Step 4) gives a different split than $\|0\rangle$ input step (Step 3)	Expected - this is physically correct if the split is not exactly 50/50 due to slight alignment imperfections. If the split is dramatically different (e.g., ALL/NONE), the polarizer rotation to 90° may not be achieving a pure $\|1\rangle$ state	Verify the $\|1\rangle$ input by temporarily removing the HWP: Port B should show ALL and Port A NONE with the polarizer at 90°. If not, adjust the polarizer.

LAB 6 NOT Gate with Half-Wave Plate

Goal

- Demonstrate how a half-wave plate at 45° functions as a quantum NOT (X) gate by fully inverting the polarization state from |0⟩ to |1⟩ and vice versa.
- Verify the self-canceling property of the NOT gate by placing two HWPs at 45° in series and confirming the output matches the original input.
- Investigate how varying the HWP angle affects brightness distribution between the |0⟩ and |1⟩ outputs of the PBS.

Theory

A NOT gate - also called the X gate in quantum computing - flips its input to the opposite state: |0⟩ becomes |1⟩, and |1⟩ becomes |0⟩. In this lab, the NOT gate is a half-wave plate rotated to 45°. The key rule: every degree you rotate the HWP rotates the polarization by two degrees. A 45° plate rotation gives a 90° polarization flip - fully inverting the state. The light is not absorbed; it is redirected. Apply the gate twice: two 90° rotations add to 180°, which for linear polarization returns you to the start. NOT × NOT = I.

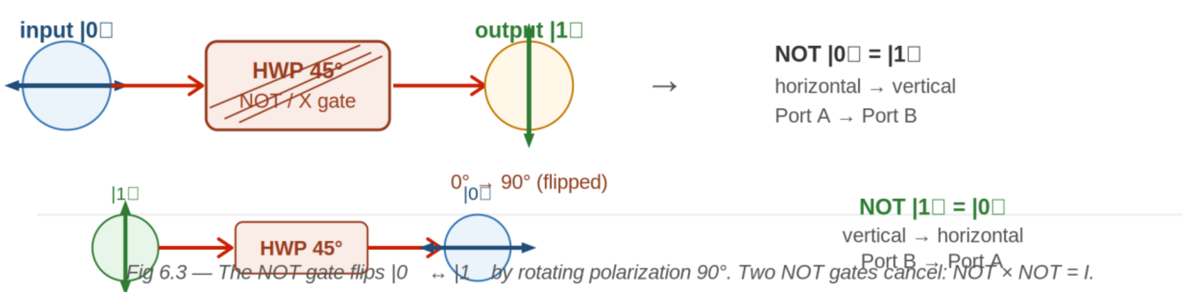

Fig 6.3 — The NOT gate flips |0⟩ ↔ |1⟩ by rotating polarization 90°. Two NOT gates cancel: NOT × NOT = I.

Equipment

- Red laser pointer (650 nm)
- Linear polarizer (always in beam path)
- Half-wave plate (HWP) - zero-order, 650 nm - two required
- Polarizing beamsplitter (PBS)
- White observation screen
- Optical board

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
HWP, first (NOT gate)	Column K, Row 6
HWP, second (NOT × NOT test)	Column L, Row 6
PBS	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

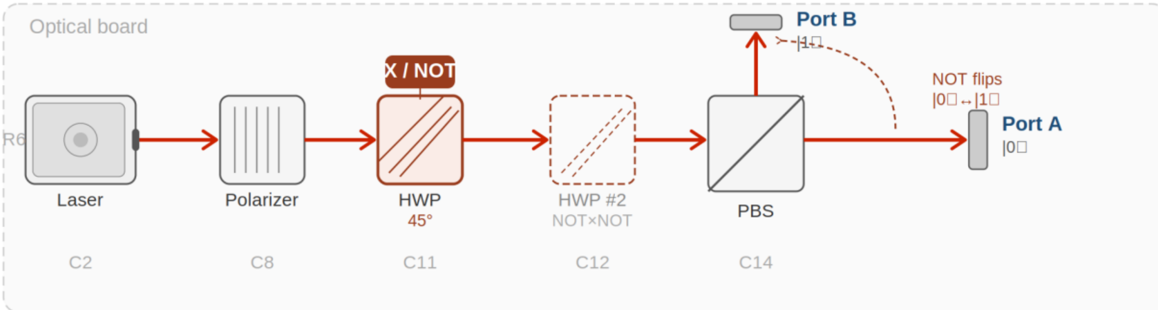

Fig 6.1 — Lab 6 beam path. HWP at 45° = NOT gate X (red). HWP #2 (dashed) tests NOT × NOT = I.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Laser and polarizer. Rotate to maximum brightness. This is ALL.

Step 2 - Establish input |0⟩

HWP at C11 at 0°. PBS at C14. Rotate polarizer until Port A is ALL and Port B is NONE.

Polarizer angle for |0⟩ input:

____

LQ1) What is the purpose of the HWP in this lab, as opposed to the polarizer?

Answer:

____

____

Step 3 - Apply NOT gate

Set HWP to 45°.

Port A brightness:

____

Port B brightness:

____

Step 4 - NOT gate on |1⟩ input

Rotate polarizer 90° to create |1⟩ input. Keep HWP at 45°.

Port A brightness:

____

____

Port B brightness:

____

____

Step 5 - NOT × NOT = I

Return polarizer to |0⟩ . Add second HWP at C12 at 45°. Predict first.

Predicted Port A:

____

____

Predicted Port B:

____

____

Observed Port A:

____

____

Observed Port B:

____

____

LQ2) Did the two HWPs cancel and return the light to |0⟩ ? What does this tell you about applying NOT twice?

I expected ___ because ___. I saw ___.

Answer:

____

____

____

Step 6 - Angle sweep

Remove one HWP. Rotate the remaining one to each angle and observe.

▼ **Visual Observation Scale** ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

HWP Angle (°)	Port A ($|0\rangle$) Brightness	Port B ($|1\rangle$) Brightness
0°		
22.5°		
45°		
67.5°		
90°		

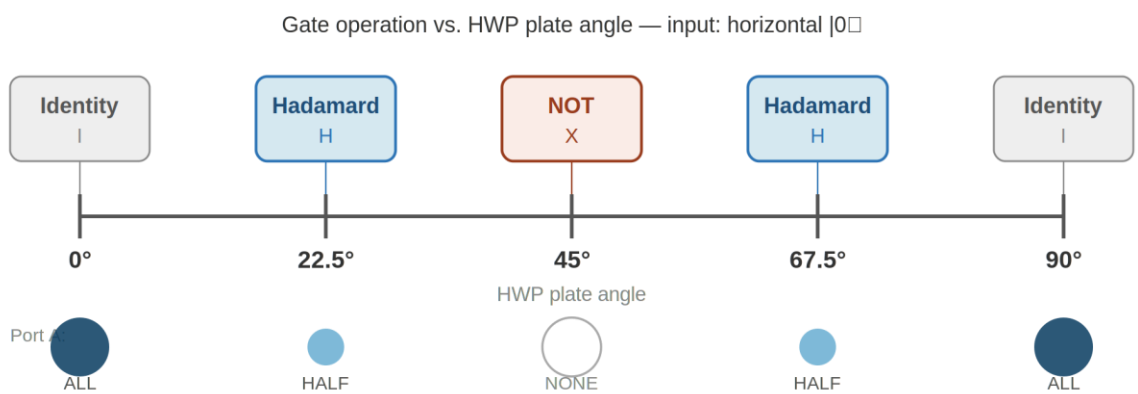

Fig 6.2 — Every gate in this course maps to one HWP angle. Circle size = Port A brightness.

Turn off the laser and put apparatus away.

Post-Lab Questions

30. What is a NOT gate? What does it do to $|0\rangle$ and to $|1\rangle$?

____

____

31. What does the HWP do to polarization, and how does a 45° rotation produce a NOT operation?

32. What happened at 22.5° in the angle sweep? Which gate from a previous lab does this remind you of, and why?

33. If a single photon in $|0\rangle$ passed through an HWP at 22.5°, what is the probability it would be measured as horizontal vs vertical? How does that compare to the brightness you saw with the laser?

34. Name two practical uses of a NOT gate - one classical, one quantum.

35. We used a laser. In one sentence, why is this still a useful model for the single-qubit X gate, and what would look different one photon at a time?

INSTRUCTOR PAGE - LAB 6: NOT GATE WITH HALF-WAVE PLATE

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Designated Laser Operator (DLO) - Lab 6

Lab 6 DLO: rotate to Student B. Lab 6 uses two HWPs simultaneously in Steps 5 and 6 - confirm both RSP05 mounts are in position at C11 and C12 before starting.

Equipment Notes - Lab 6

Equipment note: Lab 6 is the most mechanically straightforward lab in the sequence - the NOT gate is the most intuitive quantum gate, and its optical implementation is direct. The key pedagogical goal is the NOT × NOT = I result in Step 5: students who correctly predict ALL at Port A after applying two 45° HWPs in series are demonstrating genuine understanding of the double-angle rule and the self-inverse property. If students are surprised by this result, revisit the double-angle rule: 45° + 45° = 90° plate rotation → 180° polarization rotation → return to original direction.

Equipment note: The angle sweep in Step 6 connects Lab 6 to Labs 2 and 7. At 0°, Port A = ALL (identity). At 22.5°, Port A = HALF (Hadamard). At 45°, Port A = NONE (NOT). At 67.5°, Port A = HALF again. At 90°, Port A = ALL (identity again). This pattern is the $\cos^2(2\theta)$ curve reduced to five visual data points. Students who recognize 22.5° from Lab 5 are building the conceptual map between gate operations and HWP angles.

LQ Answer Key - Lab 6

LQ1 - Quick Reference: The HWP rotates the polarization direction without absorbing light. The polarizer fixes the input polarization; the HWP changes where the PBS routes the output.

Extended explanation: The distinction is the same as in Lab 2: the polarizer is a one-way measurement (projective, lossy); the HWP is a reversible rotation (unitary, lossless). In the context of quantum gates, the HWP is the gate and the PBS is the measurement. The polarizer is state preparation. Students should be able to name these three functional roles by Lab 6.

LQ2 - Quick Reference: Yes - two 45° HWPs cancel out. The first rotates polarization 90° ($|0\rangle \to |1\rangle$); the second rotates it another 90° ($|1\rangle \to |0\rangle$), returning to the original state.

Extended explanation: The Jones matrix for this: NOT × NOT = [[0,1],[1,0]] × [[0,1],[1,0]] = [[1,0],[0,1]] = I. The optical result is Port A = ALL, Port B = NONE - identical to the baseline with no HWP. This is one of the clearest demonstrations of a unitary operation and its inverse in the entire lab sequence.

Post-Lab Answer Key - Lab 6

Q1 - Quick Reference: A NOT gate inverts its input: $|0\rangle \to |1\rangle$ and $|1\rangle \to |0\rangle$.

Extended explanation: In classical computing, NOT is the fundamental one-bit operation. In quantum computing, the NOT gate (X gate) is one of the Pauli gates - it flips the qubit between the computational basis states. It is unitary (reversible), unlike a classical NOT which is also reversible but for different reasons. The optical implementation here is exact: the HWP at 45° is a perfect NOT gate for linear polarization in the H/V basis.

Q2 - Quick Reference: The HWP rotates polarization by twice its plate rotation angle. At 45°, this is a 90° polarization rotation - a perfect flip from H to V or V to H, implementing NOT.

Extended explanation: The double-angle rule is the key physical mechanism. No energy is absorbed - the light is redirected. A 45° HWP is not 'halfway' between doing nothing and doing a full rotation; it is the specific angle that produces the maximum possible polarization rotation (90°) in a single wave plate.

Q3 (22.5° in sweep) - Quick Reference: At 22.5°, Port A shows approximately HALF - the same as the Hadamard gate result from Lab 5. The HWP at 22.5° is the Hadamard gate.

Extended explanation: This connection is deliberate. The angle sweep in Step 6 is a physical scan through the space of all possible HWP operations. At 0°: identity. At 22.5°: Hadamard. At 45°: NOT. These three points on the sweep correspond to three of the most important single-qubit gates in quantum computing. The sweep also shows that there is a continuum of operations between them - not just three discrete choices.

Q4 (single photon at 22.5°) - Quick Reference: Probability 50% horizontal, 50% vertical. The laser's HALF brightness is the average of many 50/50 individual outcomes.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The new element: at 22.5° the single-photon state is $(1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$. The Born rule gives P(H) = P(V) = 0.5. With the laser, both ports show approximately equal brightness - the average of many random 0-or-1 single-photon outcomes converges to 0.5 for each port.

Q5 (uses of NOT) - Quick Reference: Classical: memory inversion, address decoding, complementary logic circuits. Quantum: state preparation (flipping $|0\rangle$ to $|1\rangle$ as input to algorithms), phase kickback in Deutsch-Jozsa, error correction circuits.

Extended explanation: Students do not need to know these applications in detail - the question is designed to broaden their sense of the gate's utility. Any two reasonable answers, one classical and one quantum-adjacent, are acceptable at this level.

Troubleshooting - Lab 6

Symptom	Likely Cause	Fix
NOT gate step: Port B does not go to ALL when HWP is at 45°	HWP mechanical zero does not correspond to the optical zero for this setup - the true NOT angle may be slightly different from 45° on the scale	Find the true NOT angle by slowly rotating from 0° until Port A reaches minimum brightness. Mark that angle as the operational NOT setting. The mechanical and optical

		zeros of the RSP05 may differ by a few degrees.
NOT × NOT step shows HALF/HALF instead of ALL/NONE	The two HWPs are not both at the same angle - one may be at the mechanical 45° and one at the optical 45°, introducing a net rotation	Use the same operational NOT angle for both HWPs (the angle found by minimizing Port A brightness, not the mechanical scale reading). Set both to this angle and retest.
Angle sweep shows unexpected pattern - brightness does not follow the expected NONE at 45°	Polarizer is not at relative zero - input polarization is not purely horizontal, so the sweep pattern is shifted	Return to Step 1 and re-establish the polarizer's relative zero. The sweep pattern is only $\cos^2(2\theta)$ when the input is purely horizontal. Any other input angle shifts the pattern.

LAB 7 Phase Shift with Wave Plates

Goal

- Observe how a half-wave plate introduces a phase shift between polarization components and describe how transmitted brightness changes with HWP angle.
- Predict brightness qualitatively at each angle using $I = I_0 \cos^2(2\theta)$ and evaluate whether observations are consistent with the formula.
- Identify the HWP angles corresponding to maximum and minimum brightness and connect those observations to the plate's effect on polarization direction.

Theory

A wave plate has two special directions - the fast axis and the slow axis. Light polarized along the fast axis travels slightly faster through the material than light along the slow axis. That speed difference builds up a phase gap: $\delta = (2\pi/\lambda)(n_e - n_0)d$, where λ is the wavelength, d is the plate thickness, and n_e and n_0 are the refractive indices for the two axes.

A half-wave plate has a retardation of exactly π radians. For linearly polarized input, the HWP rotates the polarization direction by twice the angle between the incoming polarization and the fast axis. With polarizer and analyzer both at 0° and the HWP at angle θ, Malus's Law gives: $I = I_0 \cos^2(2\theta)$. Brightness is maximum at 0° and 90°, minimum near 45°.

Equipment

- Laser (650 nm, on optical mount)
- Optical board
- Two linear polarizers (first at Column H, Row 6; second as analyzer at Column N, Row 6)
- Two aperture holders
- Half-wave plate (HWP) - zero-order, 650 nm
- Quarter-wave plate (QWP) - zero-order, 650 nm - optional extension only
- White observation screen (Column W, Row 6)

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
Half-wave plate (HWP)	Column K, Row 6
Second linear polarizer (analyzer)	Column N, Row 6
White observation screen	Column W, Row 6

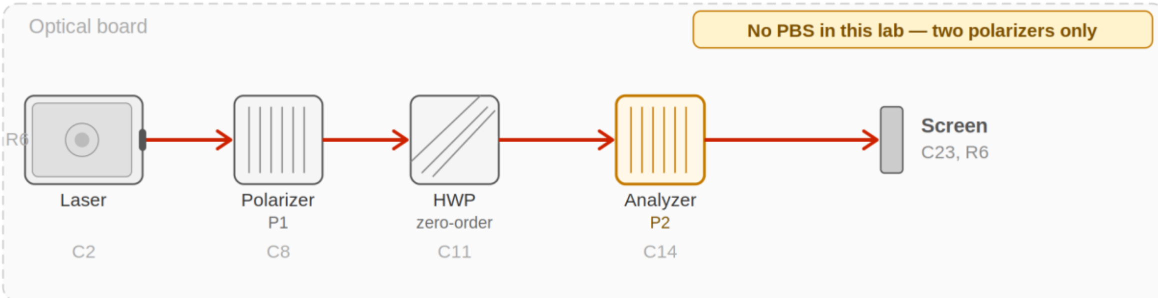


Fig 7.1 — Lab 7 beam path. Two polarizers — no PBS. The analyzer (amber) measures HWP phase output.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.
- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.
- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.
- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.
- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

Procedure

Step 1 - Set your 'ALL' reference

Polarizer at C8 and analyzer at C14, axes parallel. No wave plate. Rotate to maximum brightness. This is ALL.

Brightness without wave plate (ALL):

____

____

Step 2 - Insert HWP at 0°

Insert HWP between polarizer and analyzer. Rotate to maximum brightness. This is HWP 0°.

Brightness at HWP 0°:

____

____

LQ1) Why is brightness maximum when the fast axis aligns with the polarizer? What is the polarization state leaving the HWP?

Answer:

____

____

____

Step 3 - Find the minimum

Slowly rotate the HWP. Find the angle of minimum brightness.

HWP angle for minimum brightness:

____

____

Brightness at minimum:

____

____

LQ2) Why does brightness reach a minimum? At that angle, what is the polarization direction just before the analyzer?

The HWP rotates polarization by twice its own angle. At the minimum angle, polarization is ___ relative to the analyzer.

Answer:

____

____

____

Step 4 - Compare 0° and 90°

Return to 0°, then rotate to 90°.

Brightness at 0°:

____

____

Brightness at 90°:

____

____

LQ3) Are they approximately the same? Explain using $I = I_0 \cos^2(2\theta)$.

Answer:

____

____

____

Step 5 - Check 45°

Brightness at 45°:

____

____

LQ4) Is this close to the minimum? Explain using the formula.

Answer:

____

____

____

Step 6 - Prediction and observation table

Set HWP to 0°. Predict each level using the formula, then observe.

Guide: $\cos^2(0) = 1$ → ALL; $\cos^2(30°) \approx 0.75$ → MOST; $\cos^2(45°) = 0.5$ → HALF; $\cos^2(90°)$ → NONE.

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

HWP Angle (°)	Predicted Brightness	Observed Brightness	Match? Write one sentence.
0°			
15°			
30°			
45°			
60°			
90°			

LQ5) Where did observations match best? Where did they differ, and what might have caused any disagreement?

Answer:

____

____

____

Step 7 - Optional: Quarter-wave plate extension

Remove HWP. Insert QWP at C11 at 45° relative to the polarizer. Observe brightness through analyzer at 0°, then rotate the analyzer.

Brightness with QWP at 45°, analyzer at 0°:

____

LQ6) Does brightness change a lot or a little as you rotate the analyzer? What does this tell you about the polarization state?

Answer:

____

____

Turn off the laser and put equipment away.

Post-Lab Questions

36. What is the phase retardation of a half-wave plate in radians and degrees? What does it do to linear polarization direction?

    ____

    ____

37. Why did brightness go from maximum to minimum to maximum as you rotated the HWP from 0° to 90°?

    ____

    ____

38. When the HWP was at 0° (maximum output), was the light after the wave plate still polarized along the same direction as the polarizer? How do you know?

    ____

39. Why was brightness close to NONE when the HWP was at 45°?

40. In one or two sentences, what is retardation in a wave plate, and how does it change polarization state?

41. If a single photon passed through the HWP at 45° and then the analyzer at 0°, would it always be absorbed, always transmitted, or sometimes one and sometimes the other?

NONE brightness means almost no photons get through on average. What does 'almost none' mean for a single photon?

42. We used a laser. Why is this only an approximation of single-photon behavior?

INSTRUCTOR PAGE - LAB 7: PHASE SHIFT WITH WAVE PLATES

NOT FOR STUDENT DISTRIBUTION

Designated Laser Operator (DLO) - Lab 7

Lab 7 DLO: rotate to Student A. Lab 7 uses two polarizers (the second as an analyzer) rather than a PBS - confirm the second RSP05 mount is loaded with a polarizer, not a wave plate, before starting.

Equipment Notes - Lab 7

Equipment note: Lab 7 is the most optics-heavy lab in the sequence and the one most likely to surface the distinction between zero-order and multi-order wave plates. Zero-order plates have a true retardation of exactly $\lambda/2$ at the design wavelength (650 nm here), independent of temperature, over a broader angular range than multi-order plates. However, they are more sensitive to tilt because any deviation from normal incidence changes the effective path length through the crystal, altering the retardation. Students should see the formula $I = I_0 \cos^2(2\theta)$ hold well - if it does not, tilt is the most likely culprit.

Equipment note: The optional QWP extension in Step 7 is valuable if time permits. Circularly polarized light (QWP at 45°) should show nearly constant brightness as the analyzer rotates - a striking contrast to the $\cos^2$-varying brightness with the HWP. This visually demonstrates that circular polarization has no preferred linear direction, which connects back to Lab 4's basis-incompatibility lesson.

LQ Answer Key - Lab 7

LQ1 - Quick Reference: Brightness is maximum when the fast axis aligns with the polarizer because the HWP does not rotate the polarization in this case - the output polarization is the same as the input, passing through the analyzer at full brightness.

Extended explanation: When the HWP fast axis aligns with the incoming polarization, the light travels entirely along the fast axis - it experiences no retardation relative to any other component. The output polarization direction is unchanged (or rotated 180°, which is the same for linear polarization). The analyzer, aligned to the original polarization, transmits all of it: $I = I_0 \cos^2(0°) = I_0$.

LQ2 - Quick Reference: Brightness reaches a minimum when the HWP has rotated the polarization to 90° - perpendicular to the analyzer. The formula confirms: $I = I_0 \cos^2(2\theta\_min) \approx 0$, so $2\theta\_min \approx 90°$, meaning $\theta\_min \approx 45°$.

Extended explanation: At 45°, the HWP has rotated the polarization 90° from the input direction. The analyzer blocks this orthogonal polarization: $I = I_0 \cos^2(90°) = 0$. Students may find the minimum brightness is LITTLE rather than NONE - this reflects the finite extinction ratio of real polarizers and wave plates, not a fundamental limitation of the physics.

LQ3 - Quick Reference: Yes - brightness at 90° should be approximately the same as at 0°. The formula: $I = I_0 \cos^2(2 \times 90°) = I_0 \cos^2(180°) = I_0 \times 1 = I_0$.

Extended explanation: Rotating the HWP by 90° produces a 180° polarization rotation - which returns linear polarization to its original direction (180° and 0° are the same direction for linear polarization). The formula has a period of 90° for the HWP angle because of the double-angle factor. Students who expect the period to be 180° (like a simple polarizer) should be directed to compare the HWP formula ($\cos^2(2\theta)$) with the polarizer formula ($\cos^2(\theta)$).

LQ4 - Quick Reference: Yes - at 45°, $I = I_0 \cos^2(90°) = 0$, which is the minimum. The formula predicts NONE or close to it at this angle.

Extended explanation: This connects the formula to the observable. A student who has predicted LITTLE or NONE using the formula before rotating should find good agreement. Small deviations toward LITTLE (rather than true NONE) reflect the finite extinction ratio of the optics, not formula failure.

LQ5 (prediction vs. observation agreement) - Quick Reference: Best agreement typically at 0°, 45°, and 90° - the formula predicts ALL, NONE/LITTLE, and ALL respectively, and these extremes are easy to observe clearly. Intermediate angles (15°, 30°, 60°) may show more disagreement due to the difficulty of visually distinguishing MOST from ALL or LITTLE from HALF.

Extended explanation: The visual scale is a quantization of a continuous function. The $\cos^2(2\theta)$ curve is smooth; our five-level scale creates steps. Disagreement between prediction and observation near the inflection points of the curve (around 22.5° and 67.5°, where the curve is steep) is expected and is a good discussion point about measurement precision and quantization error.

LQ6 (QWP extension) - Quick Reference: Brightness changes very little as the analyzer rotates. This tells us the light after the QWP at 45° is approximately circularly polarized - it has no preferred linear direction.

Extended explanation: Perfectly circularly polarized light would show completely constant brightness as the analyzer rotates (the intensity is equal for all linear polarization directions). Any variation students observe reflects imperfect circularity - the QWP may be slightly off 45°, or the zero-order plate may have a small tilt. This connects directly to Lab 4: the QWP at 45° creates circular polarization, and the analyzer here plays the role the PBS played in Lab 4.

Post-Lab Answer Key - Lab 7

Q1 - Quick Reference: Phase retardation of a half-wave plate: $\delta = \pi$ radians = 180°. It rotates the direction of linear polarization by twice the angle between the input polarization and the fast axis.

Extended explanation: The formula $\delta = (2\pi/\lambda)(n_e - n_0)d$ gives the retardation in radians. For a HWP, this is engineered to equal π at the design wavelength. The physical effect: each polarization component (along fast and slow axes) accumulates different phase. After propagation, the relative phase difference between the two components is π - which for a linear input at angle θ to the fast axis produces a rotated linear output at angle 2θ past the fast axis. The net rotation of the polarization direction is 2θ (measured from the input direction). Students do not need to derive this - stating the result (π retardation, 2θ rotation) is sufficient.

Q2 - Quick Reference: The HWP rotates polarization by 2θ. At 0°, polarization is unchanged (0° rotation) → analyzer transmits fully → ALL. At 45°, polarization is rotated 90° → perpendicular to analyzer → NONE. At 90°, polarization is rotated 180° = same direction → ALL again.

Extended explanation: This is the $\cos^2(2\theta)$ pattern in words. The key insight: the period is 90° for the HWP (not 180° as for a polarizer) because of the 2θ double-angle relationship. One full cycle of the HWP takes only 90° of plate rotation. Students who have done Lab 6 should recognize 45° as the NOT gate setting - the same angle that sends all light to Port B here is the same angle that routes all light away from Port A there.

Q3 - Quick Reference: Yes - at 0° the HWP does not rotate the polarization, so the output is still along the same direction as the polarizer (horizontal). The analyzer, also at 0°, transmits it fully.

Extended explanation: The light is linearly polarized along the same direction as before the HWP. The HWP at 0° acts as a transparent plate (for this geometry) - it does introduce a phase delay, but with both components along the same axis, the net effect on polarization direction is zero.

Q4 - Quick Reference: At 45°, the polarization is rotated 90° - perpendicular to the analyzer. The analyzer blocks this polarization: $I = I_0 \cos^2(90°) = 0$.

Extended explanation: This is the same physical situation as the NOT gate in Lab 6, now viewed from the analyzer's perspective rather than the PBS's perspective. The polarization has been fully inverted relative to the measurement axis, giving minimum transmission.

Q5 (retardation) - Quick Reference: Retardation is the phase difference introduced between the two polarization components traveling along the fast and slow axes of the wave plate. This phase difference changes the polarization state from linear to elliptical, circular, or rotated linear, depending on the input angle and the magnitude of the retardation.

Extended explanation: A useful analogy: imagine two runners starting together, one on a smooth track (fast axis) and one on a gravel path (slow axis). They start in step. After some distance, the gravel runner has fallen behind - they are out of phase. The 'phase difference' is how far behind the slow runner has fallen. For a HWP, the gravel runner falls behind by exactly half a lap (π radians). This phase difference is what converts the polarization state.

Q6 (single photon at 45°, then analyzer at 0°) - Quick Reference: It would sometimes be transmitted and sometimes absorbed - with approximately 0% probability of transmission (NONE brightness ≈ 0). But due to the finite extinction ratio of real optics, a small fraction passes through.

Extended explanation: At exactly 45° HWP, the polarization is rotated 90° - perpendicular to the analyzer at 0°. For an ideal system, $P(\text{transmission}) = \cos^2(90°) = 0$. For a real system, the extinction ratio is finite (perhaps 1:1000 or better for a good polarizer), so occasionally a photon gets through. With a laser, this shows as LITTLE or NONE brightness. With single photons, the vast majority would be absorbed, and only rarely would one get through - directly reflecting the Born rule probability.

Q7 (approximation) - Quick Reference: The laser gives a continuous brightness showing the average behavior of many photons. A single photon would give a discrete detection event with probability given by $\cos^2(2\theta)$ - but each trial would be ALL or NONE, never a partial brightness.

Extended explanation: The experiment faithfully demonstrates the formula $I = I_0 \cos^2(2\theta)$ as a continuous function. With single photons, this same formula gives the probability of detection per trial. The average of many single-photon detections converges to the laser brightness. The approximation is that we observe the average directly rather than building it up from individual events.

Troubleshooting - Lab 7

Symptom	Likely Cause	Fix
Brightness at 0° and 90° are noticeably different - not both ALL	The analyzer is not parallel to the polarizer - a small angular offset shifts the entire $\cos^2(2\theta)$ curve	Remove the HWP. Rotate the analyzer to maximum brightness (parallel to polarizer). Call that the analyzer zero. Reinsert the HWP. The 0° and 90° readings should now match.
Minimum brightness is HALF rather than NONE or LITTLE	The extinction ratio of the polarizer-analyzer pair is poor, or the HWP retardation is not exactly $\lambda/2$ for the beam's wavelength	Check that the polarizers are both labeled for visible wavelengths. A polarizer optimized for a different spectral range may have poor extinction at 650 nm. Also check the HWP - confirm it is the zero-order 650 nm version.
Optional QWP step shows large brightness variation as analyzer rotates	QWP is not at exactly 45° relative to the polarizer, producing elliptical rather than circular polarization	Adjust the QWP angle slowly around 45° while rotating the analyzer and watching for the angle that minimizes the brightness variation. This is the true 45° setting for this setup.
Prediction and observation tables show systematic offset - predicted MOST, observed HALF at the same angle	The prediction was made using $\cos^2(\theta)$ instead of $\cos^2(2\theta)$ - the student is applying the polarizer formula to the HWP	Clarify the factor of two. With a polarizer, rotating by θ changes the transmission by $\cos^2(\theta)$. With an HWP, rotating by θ rotates the polarization by 2θ, so the analyzer sees $\cos^2(2\theta)$. Walk through the calculation for one angle together.

LAB 8 Sequential Gate Operations

Goal

- Demonstrate that quantum gate operations are order-dependent by comparing brightness splits for H→X and X→H sequences and observing different results.
- Describe the polarization state after each gate in a two-gate sequence and connect those descriptions to brightness at each PBS port.
- Verify the self-inverse properties of H and NOT gates by confirming that applying each gate twice returns light to original horizontal polarization.

Theory

Real quantum algorithms are long sequences of gates, and the order you choose changes the outcome. Same gates, different order, different result. The second gate always acts on whatever the first gate left behind - and the first gate changes the state differently depending on its angle.

Two optical gates: HWP at 22.5° = Hadamard (H); HWP at 45° = NOT (X). H then X is not the same as X then H. The PBS with brightness observations is a projective measurement in the H/V basis. Self-inverse properties: H then H = I; X then X = I. These let you open a transformation and close it back up precisely.

Equipment

- Laser (650 nm, on optical mount)
- Optical board
- Linear polarizer (always in beam path)
- Two half-wave plates (HWP) - zero-order, 650 nm
- Polarizing beamsplitter (PBS)
- White observation screen

Assembly - Component Positions

Component	Position
Laser	Column B, Row 6
Linear polarizer	Column H, Row 6
HWP1 - Gate 1	Column K, Row 6
HWP2 - Gate 2	Column L, Row 6
PBS	Column N, Row 6
White observation screen	Column N, Row 11 (Port B) and Column W, Row 6 (Port A)

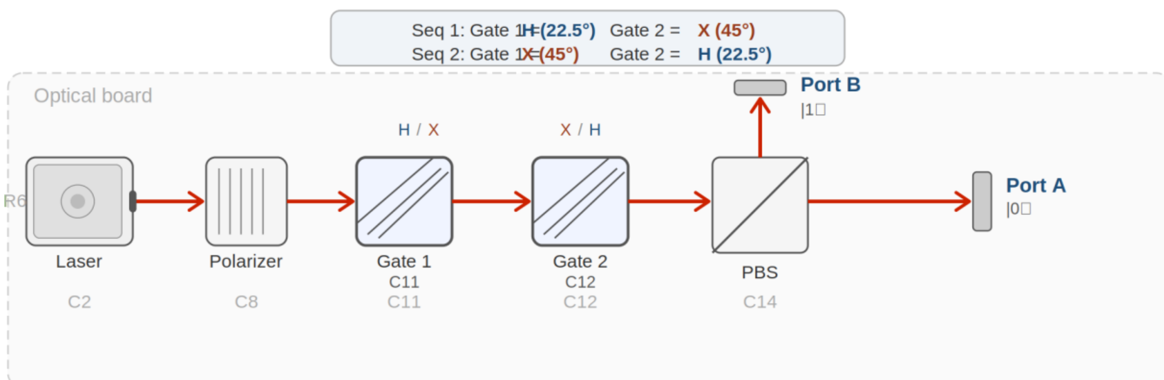


Fig 8.1 — Lab 8 beam path. Two HWPs in series. Gate assignments swap between Sequence 1 and Sequence 2.

Alignment Notes

- **Beam path row:** All components in every lab travel along Row 6 - the standard beam path row. Check that post heights are consistent so the beam travels level across the board.
- **Direction:** Align in the direction of beam propagation - left to right, column by column.

- **Beam centering:** Close the first aperture holder to a small aperture and adjust the laser or polarizer mount until the beam is centered. Repeat at the second aperture holder. Open both fully once alignment is confirmed.

- **Polarizer:** The first polarizer defines the incoming polarization and sets your 'ALL' baseline - see Step 1 of every procedure.

- **Wave plates:** Keep wave plates perpendicular to the beam. Small tilts affect the result more than you might expect with zero-order optics.

- **General:** Use a white card at each element to see the beam. Handle optics by edges or mounts only. If the beam is lost, return to the last aligned position and re-walk forward.

⚠ Safety

- **Laser radiation hazard.** Do not look into the beam or any specular reflection. Wear Class II laser safety eyewear rated for 650 nm at all times during operation. Keep the wave plate perpendicular to the beam - small tilts affect the result more than you might expect.

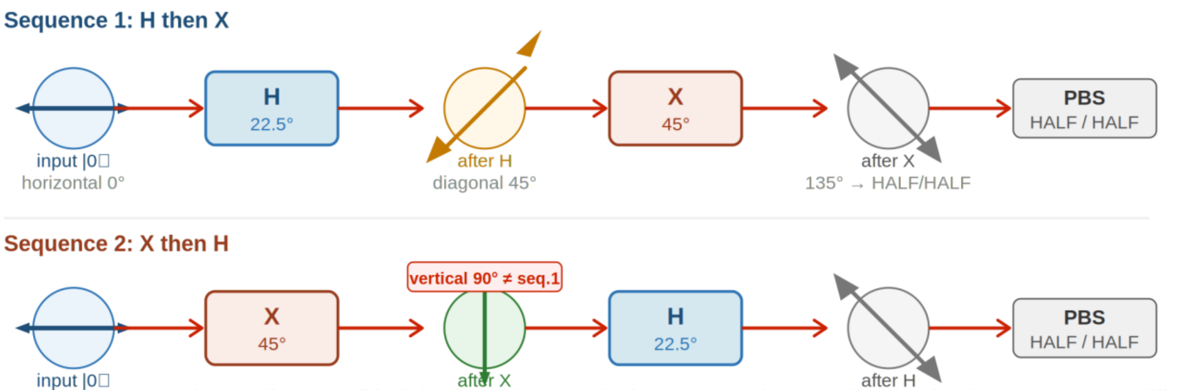
Procedure

Beam path: laser → polarizer → HWP1 → HWP2 → PBS. Polarizer provides horizontal input ($|0\rangle$).

Step 1 - Set your 'ALL' reference

No wave plates. Polarizer to maximum brightness. This is ALL.

Baseline brightness (ALL): ____



— Gate order changes the intermediate state (highlighted). For |0 input, both sequences give HALF/HALF — but the quantum states differ ir

Step 2 - Sequence 1: H then X

HWP1 to 22.5° (H). HWP2 to 45° (X). Sequence: |0⟩ → H → X → PBS. Predict first.

After H at 22.5°, polarization state is:

____

____

After X at 45°, I expect Port A to show:

____

____

I expect Port B to show:

____

____

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Predicted	Observed

Port A (|0⟩ , transmitted)		
Port B (|1⟩ , reflected)		

Total: is sum of both ports approximately ALL?:

____

____

LQ1) What did the first wave plate do, and what did the second do? How does that explain the split?

H at 22.5° rotated polarization to ___. Then X at 45° rotated it to ___. So the PBS sees ___.

Answer:

____

____

____

Step 3 - Sequence 2: X then H

Swap: HWP1 to 45° (X), HWP2 to 22.5° (H). Predict first.

After X at 45°, polarization state is:

____

____

After H at 22.5°, I expect Port A to show:

____

____

I expect Port B to show:

____

____

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Port	Predicted	Observed

Port A ($\|0\rangle$, transmitted)		
Port B ($\|1\rangle$, reflected)		

Total: approximately ALL?:

____

____

LQ2) Is the split the same as Sequence 1? If not, why does order change the result?

Sequence 1 went ___ then ___. Sequence 2 went ___ then ___. The difference is ___.

Answer:

____

____

____

Step 4 - Summary table

▼ Visual Observation Scale ▼

NONE	LITTLE	HALF	MOST	ALL

Visual Brightness Scale - use this reference for all observation tables in this lab.

Circuit	Port A ($\|0\rangle$) Brightness	Port B ($\|1\rangle$) Brightness	Total ≈ ALL?	P($\|0\rangle$) estimate	P($\|1\rangle$) estimate
H then X					
X then H					

Probability estimates: NONE ≈ 0, LITTLE ≈ 0.25, HALF ≈ 0.5, MOST ≈ 0.75, ALL ≈ 1.0.

LQ3) Is the total brightness roughly the same for both circuits? Why should that be true?

Answer:

____

____

____

Step 5 - Optional: H × H = I

Both HWPs to 22.5°. Predict first.

Predicted Port A:

____
____

Predicted Port B:

____
____

Observed Port A:

____
____

Observed Port B:

____
____

LQ4) Why does H then H return light to $|0\rangle$? What property of the Hadamard gate does this demonstrate?

Answer:

____

____
____

Step 6 - Optional: X × X = I

Both HWPs to 45°.

Observed Port A:

____
____

Observed Port B:

____
____

LQ5) Why does X then X also return to $|0\rangle$? How is this similar to H then H?

Answer:

____

____

____

Turn off the laser and put all equipment away.

Post-Lab Questions

43. What is a sequential gate operation, and why does order matter? Use H then X vs X then H as your example.

    ____

    ____

    ____

    ____

44. In $|0\rangle \rightarrow H \rightarrow X \rightarrow$ measure, what did H do, and what did X do after that?

    ____

    ____

    ____

    ____

45. In $|0\rangle \rightarrow X \rightarrow H \rightarrow$ measure, the light was vertical after the first gate. What did H do to that vertical polarization?

    ____

    ____

    ____

    ____

46. You found that total brightness was roughly the same for both circuits. Why should that be true for any sequence of wave plates?

    ____

    ____

____
____

47. When you did H then H, light returned mostly to horizontal. In quantum notation, H × H = I. Why is this property useful in designing quantum algorithms?

____
____
____
____

48. Describe a simple circuit using only H and X gates that takes horizontal light to vertical and keeps it there.

____
____
____
____

49. Real algorithms use many gates in sequence. Why is it critical to track the exact order of every gate?

____
____
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50. This lab used a laser. How would the same circuits behave with single photons, and what does your brightness observation approximate?

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INSTRUCTOR PAGE - LAB 8: SEQUENTIAL GATE OPERATIONS

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Designated Laser Operator (DLO) - Lab 8

Lab 8 DLO: rotate to Student B for the first half (Sequences 1 and 2), then to Student A for the optional steps. Lab 8 is the capstone - both students should be actively engaged in predicting, observing, and discussing results. After Lab 8, the DLO role does not need to rotate further unless additional labs are added.

Equipment Notes - Lab 8

Equipment note: Lab 8 uses two HWPs simultaneously - both RSP05 mounts at C11 and C12 must be populated before starting. Confirm that both are zero-order HWPs at 650 nm, not mixed with a QWP from a previous lab session. The HWP and QWP look nearly identical in the RSP05 mount; label the mounts or the optic edges with a small sticker dot before the session to prevent confusion.

Equipment note: The order-dependence demonstration (H then X vs X then H) is the conceptual peak of the course. Students who have internalized the double-angle rule should be able to predict both outcomes correctly before observing. The key calculation: H then X: input is |0⟩ (horizontal). H at 22.5° rotates polarization 45° → diagonal. X at 45° rotates it another 90° → 135° from horizontal. $\cos^2(135°) = 0.5$ → HALF/HALF at Port A/B. X then H: input is |0⟩. X at 45° rotates 90° → vertical. H at 22.5° rotates it 45° → 135° from horizontal. $\cos^2(135°) = 0.5$ → HALF/HALF also. Wait - both give HALF/HALF? Yes, for a horizontal input. The order matters for other inputs (e.g., starting from |1⟩). Instructors should be prepared for this potential student observation.

Equipment note: Note on the summary: for a horizontal |0⟩ input, both H then X and X then H give a 50/50 split at the PBS. The order-dependence becomes apparent if students try: (a) a |1⟩ input, or (b) a third gate in sequence, or (c) examine what gate is effectively implemented (H·X vs X·H are different matrices even if one input gives the same probability distribution). This is worth discussing explicitly as a limitation of the visual brightness measurement - identical brightness can correspond to different quantum states.

LQ Answer Key - Lab 8

LQ1 (H then X) - Quick Reference: H at 22.5° rotates polarization 45° to diagonal. X at 45° rotates it another 90° to 135°. $\cos^2(135°) = 0.5$ → approximately HALF/HALF.

Extended explanation: The sequential nature is key: the second gate acts on the output of the first, not on the original input. If students treat each gate independently of the other, they will not predict correctly. The correct chain: |0⟩ (0°) → H → 45° polarization → X → 135° polarization → PBS → $\cos^2(135°)$ at Port A = 0.5.

LQ2 (X then H) - Quick Reference: For a |0⟩ input, X then H also gives approximately HALF/HALF - the same as H then X. However, the effective combined gate is different. For a |1⟩ input, the two sequences would give different results.

Extended explanation: This is the instructionally important subtlety of Lab 8. The brightness observation does not fully distinguish the two gate sequences for a horizontal input - the probabilities happen to match. The difference is in the quantum state (specifically, the phase relationship between H and V components), which is invisible to the PBS. Instructors should use this as an opportunity to discuss the limitations of projective measurement: two different quantum states can have the same probability distribution in a given measurement basis.

LQ3 (total brightness conservation) - Quick Reference: Yes - the total brightness should be approximately ALL for both circuits. Total brightness conservation means energy is conserved: wave plates are lossless elements, so all input light must appear at one port or the other.

Extended explanation: If the total is significantly less than ALL, the most likely cause is beam clipping (some light misses both detector screens) or reflection losses at the wave plate surfaces. Zero-order plates have slightly lower reflection losses than multi-order plates, so this should be a minor effect. The conservation principle is a useful self-check: if total brightness is much less than ALL, there is an alignment problem.

LQ4 (H × H = I) - Quick Reference: Two HWPs at 22.5° cancel: the first rotates polarization 45°, the second rotates it another 45°, giving 90° total. But wait - 90° should give Port B = ALL. The resolution: the second HWP acts on the 45° diagonal polarization from the first HWP. A 22.5° plate rotation on 45° diagonal polarization gives a net rotation back to 0° (horizontal). Port A returns to ALL.

Extended explanation: The apparent paradox (two 45° rotations should give 90°, not 0°) resolves when students remember that the second HWP rotates relative to the current polarization direction, not relative to horizontal. The Jones matrix calculation makes this unambiguous: H × H = I regardless of input. At the intuitive level: the H gate 'halfway rotates' the qubit; doing it twice completes a full rotation and returns to the start.

LQ5 (X × X = I) - Quick Reference: Two HWPs at 45° cancel: the first rotates polarization 90° ($|0\rangle \rightarrow |1\rangle$), the second rotates it another 90° ($|1\rangle \rightarrow |0\rangle$). Net rotation: 180° = return to original. Port A = ALL.

Extended explanation: X × X = I is easier to intuit than H × H = I because the 90° rotations add straightforwardly. This is the same reasoning as NOT × NOT in Lab 6, now in the context of a sequential gate sequence. Both H and X are self-inverse ($H^\dagger = H$, $X^\dagger = X$) - a property shared by the Pauli X, Y, and Z gates and the Hadamard gate, but not by most other quantum gates.

Post-Lab Answer Key - Lab 8

Q1 (sequential gates, order matters) - Quick Reference: A sequential gate operation applies one gate after another to the same qubit. Order matters because each gate acts on the output of the previous gate - the state changes with each operation, so swapping the order changes what each gate 'sees' as its input.

Extended explanation: H then X: |0⟩ → (diagonal) → (135° from H) → 50/50 split. X then H: |0⟩ → (vertical) → (diagonal from V) → 50/50 split. For this specific input (|0⟩), the PBS output probabilities happen to be the same. But the effective gate (H·X vs X·H) is different, which would be apparent with a different input or a third operation. Instructors can demonstrate with a |1⟩ input: H then X on |1⟩ gives a different result than X then H on |1⟩ .

Q2 (|0⟩ → H → X) - Quick Reference: H took |0⟩ (horizontal) to an equal superposition of |0⟩ and |1⟩ (diagonal, 45° polarization). X then flipped that diagonal state to 135° from horizontal - another equal superposition, but with opposite phase.

Extended explanation: The phase distinction (45° vs 135° diagonal) is invisible to a brightness measurement but is physically real. The state after H then X is different from the state after H alone, even though both show approximately HALF/HALF brightness at the PBS. Phase is what distinguishes these states - and phase is what quantum algorithms manipulate.

Q3 (|0⟩ → X → H, what does H do to |1⟩ ?) - Quick Reference: X took |0⟩ to |1⟩ (vertical polarization). H then rotated that vertical polarization to 135° from horizontal - an equal superposition, but with a different phase than the |0⟩ → H result.

Extended explanation: This is where the order-dependence becomes real at the quantum state level. H applied to |0⟩ gives (1/√2)(|0⟩ + |1⟩). H applied to |1⟩ gives (1/√2)(|0⟩ − |1⟩). The minus sign is the phase difference. A PBS cannot see this minus sign - but an interference experiment could. At the HS/first-year level, stating that the phase is different is sufficient.

Q4 (total brightness conservation) - Quick Reference: Wave plates are lossless (unitary) elements - they redirect light but do not absorb it. All input light must exit at one of the two PBS ports. Total brightness ≈ ALL is the conservation check.

Extended explanation: If the total were much less than ALL, energy would be going somewhere - most likely to a reflection off a wave plate surface or beam clipping at a mount. Zero-order wave plates have anti-reflection coatings that minimize this. This conservation principle is the optical analogue of unitary evolution in quantum mechanics: quantum gates preserve the total probability ($|\alpha|^2 + |\beta|^2 = 1$ at all times).

Q5 (H × H = I usefulness) - Quick Reference: H is its own inverse, so applying H twice does nothing. This allows quantum algorithms to 'open' a superposition (apply H), do something inside it, and then 'close' it back (apply H again) - perfectly reversibly.

Extended explanation: Concrete examples: Deutsch algorithm (H, query, H - extracts global function property), Grover algorithm (H at start to create uniform superposition, H is not used to close but the diffusion operator uses it internally). The self-inverse property means H is both the 'prepare superposition' step and the 'undo superposition' step - one gate serves both purposes. This is computationally efficient and elegantly simple.

Q6 (circuit taking H to V and keeping it there) - Quick Reference: X: a single NOT gate. Starting from |0⟩ (horizontal), one HWP at 45° gives |1⟩ (vertical). No second gate is needed.

Extended explanation: Students may propose H then H then X, or other sequences. These also work but are longer. The simplest circuit is just X. A good extension question: what circuit takes |0⟩ to |+⟩ (the +45° diagonal) and keeps it there? Answer: H. To take it to |−⟩ (the −45° diagonal): X then H, or equivalently H then Z (though Z has not been introduced).

Q7 (why gate order matters for algorithms) - Quick Reference: Because each gate acts on the output of the previous one - swapping gates changes what state each gate receives as input, leading to a different final state and different measurement probabilities.

Extended explanation: Real quantum algorithms are designed around specific gate orders that produce desired probability distributions at the output. A single transposed gate can make the algorithm give the wrong answer. This is analogous to the difference between 'add then multiply' and 'multiply then add' in arithmetic - both use the same operations but produce different results. For quantum algorithms, the sensitivity to order is even greater because the gates interact non-commutatively.

Q8 (single photons, same circuits) - Quick Reference: With single photons, each trial produces a definite detection at Port A or Port B - not a brightness level. The brightness students observed is the probability of detection at each port, which would be confirmed by running many single-photon trials and counting.

Extended explanation: See Lab 1 Teacher Notes for the full single-photon discussion. The specific addition: the sequential gate operations are identical for single photons and laser beams - the same angles, same PBS, same port probabilities. What changes is the observable: with a laser, we read brightness directly; with single photons, we count detection events and compute relative frequencies. The count ratio converges to the brightness ratio in the limit of many photons. This is the operational meaning of 'the laser is a good approximation for the single-photon case' - it gives the same probabilities, averaged over many photons.

Troubleshooting - Lab 8

Symptom	Likely Cause	Fix
H then X and X then H give identical brightness for both |0⟩ input AND |1⟩ input	Both HWP angle references are being set relative to different zeros - the 'same' angle on each mount is not the same optical angle	Establish a common reference: set each HWP individually to its optical zero (maximum Port A brightness) and mark those positions. From that common zero, set H = 22.5° and X = 45° on each mount. Retest.
H × H step does not return to ALL at Port A - shows HALF/HALF instead	One HWP is at 22.5° from its mechanical zero, the other from its optical zero - they have different effective angles	As above: use optical zeros for both. Set each HWP alone to 22.5° from its optical zero, verify HALF/HALF individually, then place both in the beam and verify they cancel.
Total brightness for both circuits is significantly less than ALL	Beam is clipping at one of the mount apertures when two HWPs are in the beam path at adjacent positions C11 and C12	Check beam path with a white card between HWP1 and HWP2, and between HWP2 and the PBS. Adjust mount heights and lateral positions so the beam passes through the center of each aperture.
Students cannot predict the H then X result even after reviewing the double-angle rule	Students are applying the rule to each gate's rotation angle in isolation, not chaining the rotations	Walk through the chain explicitly on the board: (1) input = 0° polarization; (2) H rotates by 2×22.5° = 45°, so polarization is now at 45°; (3) X

		rotates by 2×45° = 90° from the current 45°, giving 45+90 = 135°; (4) $\cos^2(135°) = 0.5$ → HALF at Port A. Each step changes the current polarization angle, not the input angle.